Fair dispatching with price-responsive couriers: an end-of-day earnings guarantee for the meal delivery routing problem

Daniel Giraldo-Herrera[a,*], David Álvarez-Martínez[a]

[a] *Universidad de Los Andes, Cra. 1 No 18a-12, 111711, Bogotá, Colombia*

Abstract

We study dispatching problems in which a platform repeatedly assigns work to self-employed agents who may reject it, and in which the platform is accountable for how the earnings of those agents are distributed at the end of the day rather than within each decision window. Our focal application is meal delivery, where every two minutes a platform offers routes and pay to crowdsourced couriers, couriers accept or reject according to what the offer pays per kilometre, and fairness-aware dispatching has so far balanced work within the window while assuming that offers are always accepted. We formulate the problem as a constrained Markov decision process whose state carries an equity history (each courier's accumulated earnings, orders and active time), whose constraint bounds by a threshold α the end-of-day Gini coefficient of hourly earnings over the couriers connected for at least one hour, idle ones included, and whose transition draws acceptance from a logit on pay per kilometre with two empirically observed regimes, manual acceptance and auto-accept. The formulation contains corporate, workload-balance and minimum-wage dispatching as special cases. We develop three policies that solve the same problem inside one simulator against four benchmarks: **BRKGA-Fair** biases a genetic decoder by the equity deficit, **Pay-BRKGA** also sets per-offer pay within a bounded band, and **MIP-Fair** solves a per-window mixed-integer programme with a block-exact linearisation of the Gini constraint. Acceptance is calibrated from 135 intercept surveys and 44 interviews with couriers in Bogotá. Using 24 operating days from six Colombian cities, we quantify the coverage and punctuality given up for the guarantee by policy, by acceptance regime and by threshold, and select α by three criteria. The central finding is a question of population: the courier-centric dispatcher, which lets every free courier take a route with the least-served courier first, ends the day with a Gini coefficient of 0.30 among the eligible couriers and gives work to 97% of them, whereas every order-centric policy, the three proposed ones included, ends between 0.52 and 0.57 because it leaves a quarter of the eligible couriers idle; the same statistic restricted to the couriers who delivered tells the opposite story. A guarantee must therefore be defined over a population that the dispatcher cannot shrink, and a policy must read exposure, not productivity, in the equity history: reading exposure over the eligible couriers and offering the most-deprived couriers to the solver closes 45–57% of that gap with 25 extra candidates per route and, for **BRKGA-Fair**, 59–81% with 50, at no coverage cost.

Keywords: Routing, Meal delivery, Fairness, Constrained Markov decision process, Crowdsourced couriers

1. Introduction

Dispatching problems arise when a limited set of agents must be matched repeatedly to requests that appear over time. In a corporate fleet, the agents are vehicles: an assignment is executed as soon as it is made, and the quality of a dispatching policy is measured by the service it delivers to customers (Dantzig and Ramser 1959, Pillac et al. 2013). In platform operations, however, the agents are self-employed workers who decide whether to execute each assignment and who earn their living from the sum of the assignments they execute over a day. Two features of such settings are absent from the corporate model. First, an assignment is an offer: whether it is accepted depends on what it pays relative to the effort it asks for, so the transition of the system is endogenous to the pay decision. Second, the platform is accountable for a distributional outcome, the earnings of its workers at the end of the day, which no single decision window determines but every decision window affects.

Meal delivery is the focus of this work and the setting in which these features are sharpest. On-demand platforms coordinate restaurants, customers and a crowd of couriers through an assignment decision repeated every few minutes. The operational literature on the meal delivery routing problem (MDRP) has made that decision efficient: rolling-horizon re-optimisation, order bundling and anticipatory assignment deliver high coverage with short click-to-door times (Reyes et al. 2018, Yıldız and Savelsbergh 2019, Ulmer et al. 2021). The couriers' side has received less attention: a courier earns over a day, not over a two-minute window, so a dispatcher that balances work within each window can still end the day with very unequal hourly earnings when couriers connect at different times or some systematically receive the better routes. Moreover, a courier is not a vehicle: in the field, couriers reason in pay per kilometre and reject long, low-paying routes, and platforms have moved their workforce to an auto-accept regime, backed by sanctions, because manual rejection erodes coverage. Similar structures arise in ride-hailing (Lesmana et al. 2019), in crowdsourced last-mile delivery (Sobhanan et al. 2024) and in gig platforms that guarantee an hourly wage (Nair et al. 2022): a platform decides who receives which work and at what pay, workers respond to that pay, and the fairness outcome is defined over a horizon longer than the decision window.

Methodologically, these problems are challenging because the fairness guarantee, the acceptance behaviour and the routing decision are coupled within the same sequential process. The value of

*Corresponding Author
Email addresses: d.giraldoh@uniandes.edu.co (Daniel Giraldo-Herrera), d.alvarezm@uniandes.edu.co (David Álvarez-Martínez)

offering a route to a courier depends on whether the courier will accept it, which depends on the pay; the pay changes the platform's cost and the courier's earnings; and the earnings accumulated by every courier determine whether the end-of-day guarantee can still be met by the remaining windows of the day. Anticipation is therefore required, not of future demand, which rolling-horizon dispatching handles procedurally, but of the distributional consequence of the current assignment at the end of the day: who receives each route and what the offer pays must be decided window by window while keeping track of what every courier has earned and worked so far.

We address this problem as a constrained Markov decision process (CMDP, Altman 1999). The state carries, in addition to the positions of couriers and the pool of orders, an *equity history* $\mathcal{H}_t$ with the earnings, orders and active time accumulated by every courier. The constraint bounds by a tolerated inequality α the end-of-day Gini coefficient of hourly earnings over the *eligible* couriers, those connected for at least one hour, with the idle ones counted at zero. The transition draws the acceptance of every offer from a logit on pay per kilometre, estimated from field data, with a manual regime in which couriers reject and an auto-accept regime in which they do not. We then develop three policies that read $\mathcal{H}_t$ when deciding the current window: **BRKGA-Fair** biases the decoder of a biased random-key genetic algorithm by each courier's equity deficit, **Pay-BRKGA** adds a pay gene per courier that sets the offer within a bounded band around the tariff, and **MIP-Fair** solves a per-window mixed-integer programme in which the Gini constraint is written on the projected end-of-window history and linearised exactly by blocks of courier pairs. In terms of the framework of Powell (2021), all three are cost function approximations: they modify the problem solved in each window with terms computed from $\mathcal{H}_t$ instead of learning a value of the future. The three policies and four benchmarks share route candidates, decision window, acceptance model and metrics inside a single discrete-event simulator.

This study is guided by three research questions that connect the structure of the decision problem, the proposed policies, and the operational conditions under which the guarantee is affordable. First, does an end-of-day fairness guarantee require the equity history in the state, or can fairness enforced within each window deliver it? Second, which lever satisfies the guarantee at the lowest cost in coverage and punctuality, choosing who receives each route or choosing what the offer pays, and does the answer depend on the acceptance regime? Third, under which operational conditions, acceptance regime, logit threshold, tolerated inequality and fleet composition, is the guarantee cheap or expensive, and which α is sustainable for a platform? Using 24 operating days from six Colombian cities with acceptance calibrated in the field (135 surveys, 44 interviews), we compare the three policies against a corporate greedy dispatcher, a coverage ILP, a coverage-only BRKGA and a hierarchical policy that is fair within each window. The computational study establishes the following insights:

- The guarantee is a property of the day, not of the window: a policy that is fair within every window (**Hierarchical**) ends the day more unequal than the corporate dispatcher in all six cities (0.53 against 0.30), and only the policies that read $\mathcal{H}_t$ as exposure and offer the

most-deprived couriers to the solver halve that gap, at no coverage cost.

- At the scale of a 3-km dispatching zone, coverage differs by less than one point between policies, so the cost of the guarantee is paid in *who* receives each route and in lateness (0.5–1.8 minutes) rather than in orders delivered.
- The acceptance regime matters more than the dispatching policy in the manual world: with a flat tariff, long single orders are rejected with high probability, and the regime loss is 1.6–4.8 points of coverage against at most 1.2 between policies.
- The pay lever acts even where the assignment lever cannot: **Pay-BRKGA** keeps coverage with 12% (manual) to 20% (auto-accept) less total pay because the resulting pay per kilometre remains above the threshold.
- Dispatching alone cannot lift couriers to the statutory minimum wage at the current tariff (98–100% remain below it under every policy, 92% in the survey); what it can do is distribute the shortfall evenly.
- The three criteria for choosing α (historical anchor, inequality bands, Pareto knee) give the interval $[0.30, 0.45]$ rather than a point, from the status quo of the corporate dispatcher, which no order-centric policy reaches on any day, to the smallest value met on most days by a second-generation policy at no coverage cost; we recommend $\alpha^* = 0.45$.

The paper makes three contributions. First, as a theoretical and modelling contribution, we formulate meal-delivery dispatching as a CMDP with an intertemporal equity history in the state, an end-of-day Gini constraint and price-endogenous acceptance. The formulation contains the corporate model ($\alpha = 1$), workload-balance models and minimum-wage models as special cases, and we show by a numerical example why per-window fairness does not imply end-of-day fairness. Second, as a methodological contribution, we develop three cost function approximations for the same CMDP, including a block-exact linearisation of the Gini constraint that reduces the per-window programme from $O(n^2)$ pairs to a problem solvable in a fraction of a second with thousands of connected couriers, and we verify all three against exhaustive enumeration on a toy instance. Third, as a managerial contribution, we measure the cost of the fairness guarantee on 24 real operating days from six cities with field-calibrated behaviour: the frontiers per policy and regime, the loss attributable to the acceptance regime, the sensitivity to the behavioural threshold, the choice of α and the side effect on emissions. To the best of our knowledge, this is the first MDRP formulation in which fairness is guaranteed at the end of the day rather than per window and in which courier acceptance responds to the pay the platform decides.

The remainder of this paper is organised as follows. Section 2 reviews the related literature and Section 3 defines the problem. Section 4 presents the three policies, Section 5 the experimental setting and Section 6 the computational study. Section 7 synthesises the findings and Section 8 concludes; the Appendix provides additional detail on the literature, the formulation, the algorithms and the results.

2. Related work

The problem studied in this paper lies at the intersection of three literature streams: the operational stream on the MDRP (routing and assignment), the stream on fairness and compensation in delivery and ride-hailing platforms, and the behavioural stream on courier acceptance and availability. We review the three streams, which motivate our benchmark policies, and then the methodological positioning of the proposed policies; Table 1 summarises the positioning and Appendix A provides a broader review.

	Paper	Courier welfare		Courier behaviour and pay			Real data
		Fairness metric	**Guarantee horizon**	**Acceptance**	**Pay decided**	**History in state**	
MDRP operations	Reyes et al. (2018)			perfect			✓
	Yıldız and Savelsbergh (2019)			perfect			✓
	Steever et al. (2019)			perfect			
	Ulmer et al. (2021)			perfect			✓
	Liu and Luo (2023)			perfect			✓
	Auad et al. (2023)	satisfaction	shift	perfect			✓
	Simoni and Winkenbach (2023)			perfect			
	Neria et al. (2025)			perfect			
Fairness and pay	Wen et al. (2010)	workload	horizon	perfect			
	Liao et al. (2020)	workload	instance	perfect			
	Lesmana et al. (2019)	income Gini	window	perfect			✓
	Gupta et al. (2022)	hourly-income gap	day	perfect			✓
	Nair et al. (2022)	hourly wage	shift	perfect	idle pay		✓
	Martínez-Sykora et al. (2024)	orders range	shift	perfect			✓
	Sobhanan et al. (2024)	workload	instance	perfect			
Behaviour	Ahamed et al. (2021)			exogenous			
	Bozanta et al. (2022)			exogenous			
	Quintero Rojas (2020)			distance			✓
	TODO authors (2025)	[TODO:]	reward term	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	Our work	hourly-earnings Gini	day	pay per km	✓	✓	✓

Table 1: Positioning of our work relative to the literature across courier welfare, courier behaviour and pay, and data. [TODO: complete the row of TODO authors (2025) (placeholder bib entry).]

Problem-related literature. The MDRP was defined by Reyes et al. (2018) as a rolling-horizon, dynamic-deterministic problem: at each re-optimisation epoch a deterministic instance is solved and the horizon rolls forward as new orders arrive, so uncertainty is handled procedurally rather than through anticipation. Their public instance library (Grubhub) has become the standard benchmark. Yıldız and Savelsbergh (2019) obtained provably high-quality solutions for a deterministic variant and found that bundling by itself does not reduce travel much; courier capacity and shift scheduling are the main levers. Steever et al. (2019) revise assignments whenever an order arrives or traffic changes; Ulmer et al. (2021) delay assignment until preparation-time uncertainty resolves and anticipate it with approximate dynamic programming; Liu and Luo (2023) dispatch and route on-demand deliveries from stores with random demand. Later contributions address shared logistics (Bi et al. 2023), courier satisfaction through dynamic operating regions (Auad et al. 2023), order batching for crowdsourced fleets (Simoni and Winkenbach 2023) and

ghost kitchens (Neria et al. 2025). For the broader class of stochastic dynamic vehicle routing problems we refer to Soeffker et al. (2022) and Hildebrandt et al. (2023), and for their modelling as sequential decision processes to Ulmer et al. (2020). Across these models, acceptance is perfect or exogenous and couriers are treated as vehicles. This literature motivates our corporate benchmarks: the **Greedy** dispatcher reproduces the platform's practice, in which the free couriers pick work in turn and each takes its nearest feasible route, and the **ILP** is the exact per-window coverage dispatcher that our **MIP-Fair** policy extends.

Fairness has entered delivery and ride-hailing platforms mostly as *balance*. Wen et al. (2010) balance workload and income opportunities within a multi-period routing model, and Liao et al. (2020) balance workload in meal delivery with a two-stage NSGA-II, arguing that an equitable distribution is itself an incentive. Lesmana et al. (2019) define fairness as income inequality among ride-sourcing drivers and trace the efficiency–fairness frontier with a reassignment algorithm applied batch by batch. FairFoody (Gupta et al. 2022) and WORK4FOOD (Nair et al. 2022) treat fairness as income distribution among couriers in Indian food-delivery data: the former balances earnings by assignment, the latter guarantees a minimum hourly wage by controlling the number of active couriers and paying for idle time. Martínez-Sykora et al. (2024) explore fairness constraints in food-delivery routing and scheduling on UK data and find moderate efficiency costs, and Sobhanan et al. (2024) give bi-objective equity–cost frontiers for crowdsourced last-mile delivery. These works share two limits that this paper removes: fairness is enforced per window or per instance, or through a reward term whose end-of-day consequence is not guaranteed, and couriers accept whatever is assigned. The per-window logic motivates our **Hierarchical** benchmark, which equalises earnings among the couriers of each window and ignores the history; comparing it with **BRKGA-Fair** and **MIP-Fair** isolates the value of carrying $\mathcal{H}_t$ in the state. On the economic side, Reyes et al. (2018) and Yıldız and Savelsbergh (2019) include a shift-level guaranteed minimum set by the company, Bajec and Tuljak-Suban (2022) use Shapley-value incentives for sustainable crowdshipping, and the algorithmic-fairness literature shows that short- and long-horizon fairness can diverge (Corbett-Davies et al. 2023). None of these integrates pay as a dispatching decision to which acceptance responds.

Courier behaviour has been modelled mainly as availability. Reinforcement-learning approaches (Ahamed et al. 2021, Bozanta et al. 2022) include courier availability in the state but keep acceptance exogenous, and Haferkamp et al. (2024) address repositioning. The closest precedent to endogenous acceptance is the discrete-event framework of Quintero Rojas (2020), in which the probability of accepting a notification depends on the assignment distance and a courier-level parameter, but not on the pay offered. Customer acceptance has been anticipated in delivery routing (Ulmer and Thomas 2020), but there it is the platform that accepts requests, not the agent who accepts work. No MDRP study models acceptance as a function of the offered pay, although a courier offered a long, low-paying route is more likely to reject it, which biases both coverage and fairness estimates. [TODO: One sentence on what TODO authors (2025) (arXiv:2501.05808)

actually does, after reading it.]

Methodological literature. Our policies belong to the class of cost function approximations in the framework of Powell (2021) and Powell (2019): the problem solved in each window is modified by terms, the equity deficit, the projected Gini excess and the pay excess, that encode the end-of-day consequence of the current decision, without learning a value function. This is the class of parametrised policies of the dynamic routing literature (Soeffker et al. 2022), whose parameters are tuned by enumeration or learned by simulation (Stein et al. 2025); we tune ours over a grid and report the full frontier, since the trade-off itself is the object of study. Two methodological components are specific to this paper. The first is the biased random-key genetic algorithm (Gonçalves and Resende 2011) with rolling horizon that we ported from our earlier work on the MDRP (Giraldo-Herrera and TODO co-authors 2026, Giraldo-Herrera and Álvarez 2024); the decoder is the component that reads $\mathcal{H}_t$. The second is the linearisation of the Gini constraint, which in its natural form requires one variable per pair of couriers; because a courier takes at most one route per window, only the pairs of free couriers need variables, and we verify the reformulation against the all-pairs model. The Gini coefficient is used in its mean-absolute-difference form (Yitzhaki 1979). To the best of our knowledge, the proposed policies are the first to control an end-of-day inequality constraint on real meal-delivery operations with acceptance that responds to the pay the platform decides.

3. Problem definition

In this section, we define the fair meal-delivery dispatching problem and formulate it as a constrained sequential decision process. We first describe the problem and the field evidence on courier behaviour that shapes it. We then define the components of the process. An illustrative numerical example connecting the formulation to the dispatching decisions, and showing why per-window fairness does not imply end-of-day fairness, is provided in Appendix B.1.

3.1. Problem description

A platform connects a set of stores R, a set of users U and a set of couriers C over one operating day. Stores and couriers (connection time, position and vehicle) are known in advance; orders are revealed at placement time. Every two minutes, the dispatcher assigns to each available courier at most one route, where a route collects up to Q orders from a single store and delivers them to their users, and offers it with a pay. The courier accepts or rejects the offer. Accepted routes are executed (pick-up, drop-off); the orders of rejected routes return to the pool for the next window. Couriers are notified only while idle or picking up. An order is cancelled, and counted against coverage, when no feasible assignment exists before its hard service deadline. We assume that each user places at most one order per day and that a cancelled order does not re-enter the system, as in the public MDRP instances (Reyes et al. 2018).

Three aspects distinguish this problem from the corporate MDRP, and each is supported by field evidence collected with couriers of the platform in Bogotá (44 semi-structured interviews in 2020–2021 and an intercept survey of 135 couriers in May–June 2021; the full assumption-versus-evidence table is in the online supplement). First, *acceptance responds to pay per kilometre.* Couriers reason in pay per kilometre ("nobody takes 4.1 km for 3,300"); in the survey, pay (importance 5.71 on a 7-point scale) and distance (5.65) are the dominant factors in that order. A logit on ln(pay/km) fitted to 32 acceptance decisions by 11 interviewees gives an indifference threshold of 903 COP/km (95% interval [615; 1,242]) and slope 2.89 ([2.02; 5.23]). Second, *there are two decision regimes.* In the 2020 interviews no courier auto-accepted (0 of 12) and rejection was common (13 of 14); by 2021, 23 of 25 auto-accepted and 2 of 12 rejected. In the survey 64% auto-accept always or almost always; behind the shift are platform sanctions for rejecting (60% of surveyed couriers had been blocked, median 24 hours), and auto-accepting protects against them (odds ratio 2.76). In the auto-accept regime the courier's decision is whether to be connected at all, which the data record as on and off times. Third, *the tariff is flat in distance.* The per-kilometre rate estimated from the interviews is 155 COP/km with a 95% interval $[-103; 413]$ and $R^2 = 0.02$; the survey reports 2,200 COP per order outside the platform's bonus ladder. We therefore model a flat tariff and treat per-offer pay as a policy instrument, not as a description of current practice.

The evidence also fixes the fairness outcome. Median hourly earnings while connected were 2,622 COP in the survey and 3,957 COP in the interviews; 91.9% of surveyed couriers earned less than the 2021 statutory minimum wage per hour (4,368 COP), and the platform guarantees no hourly pay. The Gini coefficient of daily income among surveyed couriers is 0.39 outside the bonus ladder and 0.26 inside it. We therefore measure fairness on hourly earnings at the end of the day, normalised by connected time so that connecting late is not penalised, over the couriers connected for at least one hour, with the idle ones counted at zero so that the dispatcher cannot improve the statistic by leaving couriers without work, and we use the two survey values as anchors for the tolerated inequality. Service and decision times are taken from the same sources: a median in-store wait of 20 minutes with a purchased order (60 minutes in the stressed interview accounts) and a median decision time of 16 seconds for an offer displayed for 20 seconds. We note that the bonus ladder is a multi-day mechanism whereas our equity history is intraday, and that all behavioural evidence is from Bogotá; both are discussed as limitations in Section 8.

The goal of the platform is to find a dispatching and pay-setting policy that minimises expected undelivered orders and lateness over the day while guaranteeing that the end-of-day Gini coefficient of the eligible couriers' hourly earnings does not exceed a tolerated value α.

3.2. Sequential decision process

The problem is stochastic and dynamic: orders are revealed over the day, preparation times are uncertain, and the acceptance of every offer is a random variable whose distribution depends on the pay the platform decides. Because assignment and pay decisions are made every two minutes

and change the earnings and availability of couriers for the rest of the day, we model the problem as a sequential decision process (Powell 2021, Ulmer et al. 2020, Cuellar-Usaquén et al. 2024) with a constraint on a terminal quantity, a constrained Markov decision process (Altman 1999). In the following, we define its components: decision point and state, decisions, exogenous information and transition, cost function, and the constrained objective.

Decision point and state. The process consists of decision points $t \in T$ with $T = \{0, 1, \ldots, t_{\max}\}$, one at the start of every two-minute window of the day; we refer to the dispatching decision made at t as a *round*. At decision point t, the state S_t contains the information accessible to the dispatcher: the couriers, the orders waiting to be assigned, their preparation, and what every courier has earned and worked so far. Its components are:

- the courier information $\mathcal{L}_t$, giving for each courier $c \in C$ its position $\ell_c(t)$, status $\sigma_c(t) \in \{\text{idle, picking-up, in-transit, dropping-off, offline}\}$ and time to next availability $\eta_c(t)$,
- the set $\mathcal{O}_t$ of unassigned orders; order o is linked to a store $r_o \in R$ with location ℓ_{r_o} and to a user $u_o \in U$ with drop-off location ℓ_{u_o}, and has placement time a_o, ready time e_o and promised drop-off time d_o,
- the vector $\mathcal{P}_t$ of estimated remaining preparation times $\hat{p}_o(t)$ of the unassigned orders,
- and the equity history $\mathcal{H}_t = (h_c(t))_{c \in C}$, with

$$h_c(t) = \big(\text{earn}_c(t),\ \text{ord}_c(t),\ \text{act}_c(t)\big), \qquad c \in C, \tag{1}$$

the earnings, orders delivered and active (connected) time accumulated by courier c up to t. In summary, $S_t = (\mathcal{L}_t, \mathcal{O}_t, \mathcal{P}_t, \mathcal{H}_t)$, with $\mathcal{O}_0 = \emptyset$ and $h_c(0) = (0, 0, 0)$. The equity history is what enables dynamic fairness control, since a policy that reads $\mathcal{H}_t$ can prevent inequities from compounding across windows; it is absent from existing MDRP formulations, where the state contains the positions and plans of couriers but not what they have earned (Ulmer et al. 2020).

Courier c has on-time e_c, off-time $l_c > e_c$, initial location ℓ_c and vehicle type $v_c \in V$, $V = \{\text{walking, bicycle, motorcycle, car}\}$. Travel times are deterministic, $\tau(\ell, \ell') = \kappa \, \|\ell - \ell'\| / \nu(v_c)$, with a circuity factor κ and a vehicle-dependent speed $\nu(v_c)$; service times s_r at the store and s_u at the user are included in the duration of a route. A route s collects at most Q orders from a single store and is served by a single courier; $\text{dist}(s)$ is its length, τ_s its duration and $|s|$ the number of orders it contains.

Decisions. At each decision point t, the dispatcher decides which available courier receives which route and what the offer pays, a set of triples

$$x_t = \big\{(c, s, \pi_{c,s}) : c \in C_t^{\text{free}},\ s \in \mathcal{S}_t\big\}, \tag{2}$$

where C_t^{free} is the set of available couriers, $\mathcal{S}_t$ the set of feasible routes that can be built from $\mathcal{O}_t$, and $\pi_{c,s}$ the pay offered for the pair. A route is feasible for a courier if every order can be picked

up after its ready time and dropped off before its hard deadline $d_o + \Delta^{\max}$, given the courier's position and speed. A decision $x_t \in \mathcal{X}(S_t)$ is feasible if each courier receives at most one route and each order appears in at most one route.

The pay offer follows the platform's tariff, an affine function of route distance,

$$\pi_{c,s} = \pi_0 \cdot |s| + \pi_1 \cdot \mathrm{dist}(s), \tag{3}$$

with π_0 a base fee per order and π_1 a per-kilometre rate; following the field evidence of Section 3.1, the baseline tariff is flat ($\pi_0 = 2{,}200$ COP, $\pi_1 = 0$). Pay-setting policies may scale the offer by a multiplier $m_c \in [1-\delta, 1+\delta]$ around the tariff, not set it arbitrarily.

Acceptance, exogenous information and transition. After decision x_t in state S_t, two kinds of information are revealed. The first is the acceptance of every offer: courier c accepts route s offered at pay $\pi_{c,s}$ with probability

$$p_{c,s}(\pi_{c,s}) = \begin{cases} \sigma\big(b\,[\ln(\pi_{c,s}/\,\mathrm{dist}(s)) - \ln\theta]\big) & \text{manual regime,} \\ 1 & \text{auto-accept regime,} \end{cases} \tag{4}$$

where σ is the logistic function, θ the pay per kilometre at which acceptance is even and b the slope on log pay per kilometre; this is the form the field evidence supports, in which the pay decided by the platform enters the decision directly, unlike the distance-only acceptance of Quintero Rojas (2020). We denote by $\mathbf{b}_t \in \{0,1\}^{|x_t|}$ the vector of Bernoulli acceptance realisations. The second kind of information is exogenous to the decision: $\omega_{t+1} = (\Delta\mathcal{O}_{t+1}, \Delta\mathcal{P}_{t+1}, \Delta\mathcal{A}_{t+1})$ reveals the orders placed during the window, the realised preparation times, and the courier log-ins and log-outs. The transition function $\mathcal{T}(S_t, x_t, \mathbf{b}_t, \omega_{t+1})$ leads to the new state S_{t+1}:

- accepted routes are executed: the courier's status, position and time to availability in $\mathcal{L}_{t+1}$ follow from the route's pick-up and drop-off sequence, and the orders of accepted routes leave $\mathcal{O}_{t+1}$,
- the orders of rejected routes remain in $\mathcal{O}_{t+1}$, together with the new orders $\Delta\mathcal{O}_{t+1}$; orders whose hard deadline can no longer be met are cancelled,
- preparation estimates are replaced by realisations where revealed, $\mathcal{P}_{t+1}$,
- couriers that log in or out change status according to $\Delta\mathcal{A}_{t+1}$,
- and the equity history is updated for every accepted pair: $\mathrm{earn}_c(t{+}1) = \mathrm{earn}_c(t) + \mathrm{pay}(c, s, \pi_{c,s})$ and $\mathrm{ord}_c(t+1) = \mathrm{ord}_c(t) + |s|$ when the route is completed, while $\mathrm{act}_c(t+1) = \mathrm{act}_c(t) + 2$ minutes for every connected courier, accepted or not.

Cost function. The cost $\mathcal{C}(S_t, x_t, \mathbf{b}_t, \omega_{t+1})$ of a decision penalises orders that are cancelled during the window and the lateness of orders delivered during the window,

$$\mathcal{C}(S_t, x_t, \mathbf{b}_t, \omega_{t+1}) = w_1 \sum_{o \in \mathcal{O}_t} \mathbb{I}[o \text{ cancelled}] + w_2 \sum_{o \text{ delivered}} \max\big(0,\ \tau_{\text{drop}}(o) - d_o\big), \tag{5}$$

where $\mathbb{I}(\cdot)$ is the indicator function and $w_1, w_2 > 0$ weigh coverage against punctuality. Its extension with a payment term, used by the pay-setting policy, is

$$\mathcal{C}^+(S_t, x_t, \mathbf{b}_t, \omega_{t+1}) = \mathcal{C}(S_t, x_t, \mathbf{b}_t, \omega_{t+1}) + \lambda \sum_{(c,s) \in x_t} b_{c,s}\ \text{pay}(c, s, \pi_{c,s}), \tag{6}$$

where $\text{pay}(c, s, \pi_{c,s})$ is the amount paid for an accepted route and $\lambda > 0$ trades platform cost against coverage. In our experiments coverage and lateness are reported separately; the lexicographic reading coverage $\succ$ lateness used by all policies corresponds to $w_1 \gg w_2$. [TODO: state the values of w_1 and w_2 if a scalar cost is reported.]

Fairness metrics and constrained objective. Let $\text{earn}_c(T)$ be the earnings of courier c at the end of the day and $\text{conn}_c(T)$ the time it was connected, so that $w_c = \text{earn}_c(T)/\text{conn}_c(T)$ are its hourly earnings, zero if it delivered nothing. The population over which fairness is measured is the set of *eligible* couriers $C^e = \{c \in C : \text{conn}_c(T) \geq 1 \text{ h}\}$, fixed by the couriers' connection decisions and not by the dispatcher. The primary fairness metric is the Gini coefficient of hourly earnings over C^e,

$$F_{\text{Gini}}(\mathbf{w}) = \frac{\sum_{c \in C^e} \sum_{c' \in C^e} |w_c - w_{c'}|}{2\,|C^e| \sum_{c \in C^e} w_c}, \qquad \mathbf{w} = (w_c)_{c \in C^e}, \tag{7}$$

with $F_{\text{Gini}} \in [0, 1]$ and 0 perfect equality. Normalising by connected time makes Eq. (7) an exposure-adjusted statistic, and the two restrictions on the population make it a guarantee: a policy cannot lower it by leaving couriers idle, since they enter with $w_c = 0$, nor by dispatching single orders to couriers who connect for a few minutes, since they are excluded. The same statistic over the couriers who delivered at least one order, $F^{\text{act}}_{\text{Gini}}$, is the one used by the fairness literature of Section 2; it can be lowered by both manoeuvres, and Section 6.1 shows that it is. Three secondary metrics are reported over the active couriers: max–min fairness $F_{\text{MaxMin}} = \min_c w_c$, order-count balance F_{Orders}, the Gini coefficient of orders delivered per connected hour, and the minimum-wage deficit F_{Deficit}, the share of active couriers below the statutory minimum hourly wage w^*.

A solution is a policy $\pi \in \Pi$ that assigns a decision $\mathcal{X}^\pi_t(S_t)$ to each state; the optimal policy

minimises the expected cumulative cost subject to the end-of-day guarantee:

$$\pi^* = \operatorname*{argmin}_{\pi \in \Pi} \mathbb{E}\left[\sum_{t \in T} \mathcal{C}\big(S_t, \mathcal{X}_t^\pi(S_t), \mathbf{b}_t, \omega_{t+1}\big) \,\Big|\, S_0\right] \tag{8}$$

$$\text{s.t. } \mathbb{E}\big[F_{\text{Gini}}(\mathbf{w}^\pi) \,\big|\, S_0\big] \le \alpha, \tag{9}$$

$$\mathcal{X}_t^\pi(S_t) \in \mathcal{X}(S_t) \qquad \forall t \in T, \tag{10}$$

where $\alpha \in [0, 1]$ is the tolerated inequality and $\mathbf{w}^\pi$ the end-of-day hourly earnings induced by π; the day is finite, so no discounting is applied. Eq. (9) is deliberately written on an end-of-day quantity, not per decision point, which is what the equity history makes controllable; in the computational study we report, besides the mean of F_{Gini}, the fraction of realisations in which $F_{\text{Gini}} \le \alpha$ holds, the operational reading of the guarantee. A minimum-wage constraint $\mathbb{E}[w_c^\pi] \ge w^*$ is not imposed because 92% of couriers earn below w^* in the field data, so it would be infeasible on every instance; it is pursued through F_{Deficit} by the pay-setting policy (Section 4.2) and reported for every policy.

The formulation contains three familiar models as special cases: with $\alpha = 1$ it reduces to the corporate, efficiency-only MDRP of Reyes et al. (2018); with F_{Orders} in place of F_{Gini} in Eq. (9) it gives the workload-balance models of Wen et al. (2010) and Liao et al. (2020); and with a bound on F_{Deficit}, or the minimum-wage constraint alone, the wage-guarantee model of Nair et al. (2022). Since the vehicle type of every courier is recorded, the emissions of a decision, $E(x_t) = \sum_{(c,s) \in x_t} b_{c,s}\, \epsilon(v_c)\, \text{dist}(s)$ with $\epsilon(v)$ a per-kilometre emission factor (zero for walking and bicycles), are a diagnostic reported alongside the frontiers, not a constraint.

Even in the two-courier example of Appendix B.1 the challenge becomes evident: a policy that equalises the pay offered within every window can end the day with a Gini coefficient of 0.5 between two equally connected couriers where 0 was attainable at the same coverage, because the per-window statistic ignores what each courier has already earned. A policy therefore needs $\mathcal{H}_t$ to control Eq. (9); the next section presents three policies that read it.

4. Method

In this section, we present our policies. We first give a motivation and overview of the shared architecture and then present the individual policies in detail.

4.1. Motivation and overview

Effective decision making is challenging for two reasons.

- **Search:** A round decides, for up to a few hundred free couriers and a few hundred ready orders, which courier receives which of the candidate routes (singletons and bundles) and what the offer pays. The assignment part alone is a large matching problem that must be solved within the two-minute window, and the pay part adds a continuous decision per pair.

- **Evaluate:** The value of a decision has two components that are revealed at different times. Whether a route is accepted is revealed within seconds and depends on the pay; whether the end-of-day guarantee is met is revealed only at the end of the day and depends on every subsequent round. The second component cannot be evaluated per window: as the example of Appendix B.1 shows, two decisions that are indistinguishable by any statistic of the current window can lead to end-of-day Gini coefficients of 0 and 0.5.

We address these challenges as follows. All policies, proposed and benchmark, are implemented as the assignment step of one discrete-event simulator and share (i) a round every two minutes; (ii) the same candidate routes, namely every singleton and one bundle per store with its Q most urgent ready orders, filtered by route feasibility and by a prefilter of the k nearest available couriers per route; (iii) the acceptance model of Eq. (4) and the flat tariff; and (iv) the same metrics, computed on the end-of-day history $\mathcal{H}_T$. Let K_t denote the set of candidate (courier, route) pairs at round t. Each policy returns a subset of K_t with at most one route per courier and one route per order, together with a pay offer per pair, that is, a feasible decision x_t in the sense of Eq. (2). The shared architecture is what makes the frontiers of the three policies comparable with each other and with the benchmarks.

The three policies differ in how they use $\mathcal{H}_t$ to steer the current round towards the end-of-day guarantee, and in which lever they move. **BRKGA-Fair** keeps the population and fitness of a coverage-maximising genetic algorithm and biases its decoder: among decodings of equal coverage, routes are redirected to couriers whose hourly earnings so far fall below the mean, by a weight w. **MIP-Fair** reads the constraint literally: in every round it solves a mixed-integer programme that maximises coverage first and then minimises the amount by which the projected end-of-round Gini coefficient exceeds α, with the constraint linearised exactly. **Pay-BRKGA** moves the other lever: it keeps the decoder of **BRKGA-Fair** without the deficit bias and adds one pay gene per courier, so that the offer can rise for couriers below the minimum wage and fall where acceptance is not at stake, within a band around the tariff; its fitness trades expected coverage, projected minimum-wage deficit and pay excess with a weight λ_{pay}. In terms of the framework of Powell (2021), the three are cost function approximations: the per-round problem is modified by terms computed from the state, the deficit, the projected Gini excess and the pay excess, rather than by a learned value of the future. We do not learn these terms because the quantity they anticipate, the end-of-day Gini coefficient, is a deterministic function of $\mathcal{H}_t$ and of the remaining decisions, and the parameters w, α and λ_{pay} are precisely the knobs whose frontier the platform needs to see. Algorithm 1 summarises a round for the three policies; the following sections detail each component.

4.2. Detailed description of the policies

In this section, we provide details of every component. We first describe the coverage-maximising BRKGA that **BRKGA-Fair** and **Pay-BRKGA** extend, then the three policies, and finally their

Algorithm 1 Dispatching round shared by the three proposed policies

for $t = 0, \ldots, t_{\max}$ **do**
  build the candidate routes $\mathcal{S}_t$ (singletons and one bundle per store) from the ready orders in $\mathcal{O}_t$
  build the candidate pairs K_t by route feasibility and the k-nearest prefilter (second generation: plus the k_d free couriers with the largest deficit, the nearest among ties)
  compute from $\mathcal{H}_t$ (Eq. (1)) the hourly earnings so far $w_c(t)$ of every connected courier
  if BRKGA-Fair or **Pay-BRKGA then**
    run the BRKGA for G generations with decoder of Algorithm 2 and score of Eq. (11) (fitness of Eq. (15) for **Pay-BRKGA**)
  else if MIP-Fair then
    solve the three-stage programme of Eqs. (12) - (14) over K_t
  offer the selected pairs with pay $\pi_{c,s}$; draw acceptance from Eq. (4)
  execute accepted routes, return rejected orders to the pool, update $\mathcal{H}_{t+1}$

verification.

Coverage BRKGA and decoder. The biased random-key genetic algorithm (Gonçalves and Resende 2011) with rolling horizon is ported from our earlier work on the MDRP (Giraldo-Herrera and TODO co-authors 2026). The chromosome has one key per ready order and one per available courier. The decoder (Algorithm 2) visits stores in the order of their highest-key order; at each store it takes the head order, evaluates its candidate routes (singleton and bundle) against the K highest-key couriers among those that can feasibly serve the route, and keeps the pair with the best score. Fitness is coverage, with mean travel time as a tie-breaker. Two decoder scores are available: $\text{score}_{\text{finish}}(s, c)$, the time at which the courier finishes the route, under which a bundle never beats the singleton with the same head order, and $\text{score}_{\text{fpo}}(s, c) = \text{score}_{\text{finish}}(s, c)/|s|$, the travel time per order, under which bundles compete. The paper reports the latter so that the comparison with the **ILP** and the **Hierarchical** policy, which bundle, is like for like; the original rule is an ablation in Appendix C. We note that the decoder selects the K couriers among the candidates of the route, not among all free couriers; the earlier variant, which selected among all free couriers and then looked them up in the prefilter, collapsed at the scale of Bogotá (coverage 0.16 against 0.44 for **Greedy**) because with more than two thousand free couriers the two sets rarely intersected.

BRKGA-Fair. **BRKGA-Fair** is the coverage BRKGA with a decoder that reads the equity deficit from $\mathcal{H}_t$. Only the score changes:

$$\text{score}_{\text{fair}}(s, c) = \text{score}(s, c) - w \cdot \text{deficit}_c, \qquad \text{deficit}_c = \frac{\max(0,\ \bar{w}_t - w_c(t))}{\bar{w}_t} \in [0, 1], \tag{11}$$

where $w_c(t) = \text{earn}_c(t)/\,\text{act}_c(t)$ is the courier's hourly earnings so far (zero if no active time yet), $\bar{w}_t$ is the mean over the free couriers of the round, and w is a weight in minutes: a courier with full deficit is treated as if every route of theirs finished w minutes earlier. The fitness that ranks

Algorithm 2 BRKGA decoder shared by **BRKGA**, **BRKGA-Fair** and **Pay-BRKGA**

Input: keys κ_o for ready orders, κ_c for available couriers, candidate pairs K_t, score function $\text{score}(s, c)$

Output: assignment A

$A \leftarrow \emptyset$; mark all couriers free
for each store r in decreasing order of $\max_{o:r_o=r} \kappa_o$ **do**
  for each order o of r in decreasing order of κ_o, not yet assigned **do**
    $\mathcal{S}_o \leftarrow$ candidate routes headed by o (singleton and the store bundle containing o)
    $\mathcal{C}_o \leftarrow$ the K free couriers with highest κ_c among $\{c : (c, s) \in K_t,\ s \in \mathcal{S}_o\}$
    if $\mathcal{C}_o \neq \emptyset$ **then**
      $(s^*, c^*) \leftarrow \arg\min\{\text{score}(s, c) : s \in \mathcal{S}_o,\ c \in \mathcal{C}_o,\ (c, s) \in K_t\}$
      $A \leftarrow A \cup \{(c^*, s^*)\}$; mark c^* busy and the orders of s^* assigned
return A

the population is unchanged, so the deficit redirects routes among decodings of equal coverage and the population retains its maximum-coverage members. With $w = 0$ the policy is identical to the coverage BRKGA for the same seed, which isolates the value of the deficit term. The hourly basis is chosen for coherence with Eq. (7); a total-earnings basis would penalise connecting late rather than being under-served, and is reported as an ablation.

MIP-Fair. **MIP-Fair** solves, at each round, a mixed-integer programme over K_t with the Gini constraint of Eq. (9) written on the projected end-of-round values of $\mathcal{H}_t$. Let $x_{cs} \in \{0, 1\}$ indicate that pair $(c, s) \in K_t$ is dispatched and $y_o \in \{0, 1\}$ that order o is covered. Because a courier takes at most one route per round, the projected hourly earnings of a free courier are linear in x:

$$w_c = w_c^0\Big(1 - \sum_s x_{cs}\Big) + \sum_s x_{cs}\, w_c(s), \qquad w_c(s) = \frac{\text{earn}_c(t) + \text{pay}(s)}{(\text{act}_c(t) + \tau_s)/60}, \tag{12}$$

where τ_s is the route's travel-plus-service time in minutes. Busy and recently connected couriers enter with fixed values $w_c = w_c^0$. The Gini constraint is evaluated over all connected couriers, not only the free ones of the round, because Eq. (9) is over C^e and the free couriers are a biased subset. Two readings of $\mathcal{H}_t$ are evaluated in Section 6.1. In the first generation w_c is the earnings per *active* hour (minutes in motion or serving), a productivity rate that is nearly equal across couriers, over every connected courier; in the second generation it is the earnings per hour connected so far, the exposure basis of Eq. (7), over the eligible couriers, and the candidate set of each route adds the k_d free couriers with the largest deficit (the deficit-aware prefilter of Algorithm 1), so that an idle courier can be chosen. The same two readings apply to the deficit of **BRKGA-Fair** and **Pay-BRKGA**. Writing $n = |C_t|$ for the number of connected couriers, the constraint $F_{\text{Gini}}(\mathbf{w}) \leq \alpha$ is equivalent to $\sum_{i<j} |w_i - w_j| - \alpha\, n \sum_c w_c \leq 0$, and since the inequality is in the direction that

allows it, it is linearised exactly as

$$\sum_{i<j} d_{ij} - \alpha\, n \sum_{c} w_c \;\le\; n\, W_{\text{ref}}\, v, \qquad d_{ij} \ge w_i - w_j,\;\; d_{ij} \ge w_j - w_i,\;\; v \ge 0, \tag{13}$$

where v is an elastic, dimensionless "Gini excess" scaled by a reference W_{ref} so that no round is infeasible, and $v^* > 0$ measures by how much α could not be met in that round. Building d_{ij} for all pairs is $O(n^2)$ per round, which is prohibitive with thousands of connected couriers: with 188 connected couriers the all-pairs model did not finish two hours of Pereira in four minutes, and Bogotá would require 3.5 million pairs per round. The pairs are therefore handled in three blocks. Fixed–fixed pairs contribute a constant. Free–fixed pairs are linear in x, because the free courier's w_i takes one of finitely many values (w_i^0 or $w_i(s)$), so $|w_i - w_j| = (1 - \sum_s x_{is})\,|w_i^0 - w_j| + \sum_s x_{is}\,|w_i(s) - w_j|$. Only free–free pairs require the variables d_{ij}. The reformulation is exact, which we verify against the all-pairs model on a toy instance, and reduces the round from minutes to a fraction of a second on the medium cities. The programme is solved lexicographically in three stages, each fixing the previous optimum:

$$\max \sum_{o} y_o \;\succ\; \min v \;\succ\; \min\Big(\lambda_{\text{tt}} \sum_{(c,s)} \tau_s x_{cs} + \mu \sum_{o} \text{late}_o\Big), \tag{14}$$

subject to each order in at most one dispatched route, each courier in at most one, $y_o \le \sum_{(c,s)\ni o} x_{cs}$, and Eqs. (12) - (13). The complete formulation is given in Appendix B.2. Coverage is placed first because the alternative order (equity first) collapses from a cold start: when every courier has earned nothing, any assignment raises the projected Gini coefficient, so "assign nothing" minimises v in every round and the system never starts (coverage 0.000 in the pilot). With coverage first, the Gini excess decides *who* receives each route among the maximum-coverage assignments, which is the equity lever; $\alpha = 1$ recovers the coverage **ILP** and $\alpha \to 0$ minimises the projected Gini coefficient in every round. The equity-first variant is reported as an ablation.

Pay-BRKGA. **Pay-BRKGA** decides who is paid how much, not how routes are built. The chromosome of the coverage BRKGA is extended with one pay gene per free courier, mapped to a multiplier $m_c \in [1-\delta, 1+\delta]$, and the offer is $\pi_{c,s} = \pi_0 \cdot |s| \cdot m_c$; under the affine tariff of Eq. (3) the multiplier would apply to π_1 only. The decoder is Algorithm 2. The fitness is the scalarised objective, in coverage-fraction units and to be minimised,

$$\Big(1 - \frac{\mathbb{E}[\text{covered}]}{|\mathcal{O}_t|}\Big) + \lambda_{\text{def}}\, F^{\text{proj}}_{\text{Deficit}} + \lambda_{\text{pay}}\Big(\frac{\sum \text{pay}}{\sum \text{tariff}} - 1\Big) + 10^{-3}\,\frac{\bar{\tau}}{20}, \tag{15}$$

where $\mathbb{E}[\text{covered}] = \sum_{(c,s)\in A} |s|\; p_{c,s}(\pi_{c,s})$ uses the acceptance model of Eq. (4), so that raising pay buys coverage only if the behavioural model says it does (in the auto-accept regime pay moves only F_{Deficit} and cost); $F^{\text{proj}}_{\text{Deficit}}$ is the fraction of free couriers whose projected hourly earnings after the

round, assuming acceptance, remain below w^*; the third term is the relative pay excess over the tariff, the λ of Eq. (6); and the last term breaks ties by mean travel time $\bar{\tau}$. The scalarisation is a weighted sum rather than an explicit Pareto set because a round yields one decision; sweeping λ_{pay} traces the cost-of-fairness curve. We note that the mechanism **Pay-BRKGA** acts on is distance, not pay: under the flat tariff, a 2-km singleton pays 1,100 COP/km (acceptance about 0.64 at $\theta = 903$) while a 4-km one pays 550 COP/km (about 0.19), so the pay gene is the only lever that can make a long route acceptable in the manual regime.

Verification. The four pieces (BRKGA port, **BRKGA-Fair**, **MIP-Fair**, **Pay-BRKGA**) are covered by unit tests against exhaustive enumeration on a toy instance of three couriers and four orders: the candidate sets coincide with those of the **ILP** and the **Hierarchical** policy (18 pairs); the $\text{score}_{\text{fpo}}$ decoder reaches the exhaustive optimum of its fitness; **MIP-Fair** with $\alpha = 0$ coincides with the exhaustive lexicographic optimum under both priorities and both earnings bases, and with $\alpha = 1$ covers 4 of 4 orders like the **ILP**; the block linearisation reproduces the all-pairs model; and for **Pay-BRKGA**, with λ_{pay} alone all offers fall below the tariff, with λ_{def} alone the under-paid courier receives an offer above the tariff and leaves the deficit, and expected coverage equals coverage times the acceptance rate under a fixed-probability model. The suite has 84 tests.

5. Experimental setting

In the following, we outline the instances, the benchmark policies, the experimental design and the implementation details of our computational study.

5.1. Instances

We test our policies on 24 full operating days (00:00–24:00) exported by a leading on-demand delivery platform operating in Colombia [TODO: confirm whether the platform may be named]. Each day records every order (store, drop-off coordinates, placement, preparation and promised drop-off times) and every courier (vehicle, connection point, on and off times). Projecting the coordinates showed that the 24 days cover six cities: Bogotá, Medellín, Cali, Barranquilla, Bucaramanga and Pereira (four days each, except three for Cali and five for Bucaramanga). Full-city days range from 540 orders and 400 couriers (Pereira) to 59,000 orders and 16,800 couriers (Bogotá); fleet composition differs sharply, from 45% bicycles in Bogotá to 90% motorcycles in Pereira and Bucaramanga, which matters for the emissions diagnostic. The behavioural parameters come from the interviews and the survey described in Section 3.1; the remaining parameters are either platform practice or declared assumptions. The simulator, the policies, the instance loader and the batch scripts are available at https://github.com/Danielgiraldosb/mdrp-simulator [TODO: confirm whether the instance data can be shared; otherwise state the access conditions].

Layout. A full city day is not the problem a dispatcher solves, so the experimental unit is a *zone*: the orders (by drop-off) and couriers (by connection point) within 3 km of the instance's store centroid, over the full day. Three kilometres keeps Pereira above a few hundred orders, so that end-of-day Gini coefficients are not noise, and Bogotá below the scale at which the per-round MIP exceeds the budget; it is also of the order of the prospecting radius of the framework of Quintero Rojas (2020). Each day is projected on a plane centred on the store centroid, discarding a few points hundreds of kilometres away. Table C.6 in Appendix C.1 lists the 24 instances; zone sizes range from 342 to 10,215 orders (mean 3,612), and Figure 1 shows the zone of instance 0 in Bogotá with its stores, drop-offs and courier connection points.

Couriers and travel. Travel time is Euclidean distance times a circuity factor of $\kappa = 1.3$, standard for an urban street network relative to the straight line (1.0 in a sensitivity run), divided by the vehicle speed: 15 km/h for bicycles, 25 for motorcycles, 5 for walking and 20 for cars, declared as assumptions of the same order as the 320 m/min of the public MDRP instances (Reyes et al. 2018). Couriers connect and disconnect at the recorded on and off times; repositioning of idle couriers is switched off because most surveyed couriers wait in place. A courier decides on an offer after a delay of 16 ± 8 seconds.

Orders and service. In-store service takes 20 minutes with a purchased order (60 minutes in a stress run) and drop-off service 4 minutes, the default of the public instances. The hard deadline is the promised drop-off time plus $\Delta^{\max} = 30$ minutes, the slack the interviews describe as tolerated. Route capacity is $Q = 3$. Customer cancellations are switched off because there is no field evidence on them.

Tariff, acceptance and wage. The flat tariff pays $\pi_0 = 2{,}200$ COP per order (survey, $n = 132$); the pay band of **Pay-BRKGA** is $\delta = 0.30$. The logit of Eq. (4) uses $\theta = 903$ COP/km and $b = 2.89$, the model estimated on the 32 interview decisions under the 2020 manual regime: acceptance is 0.57, 0.81 and 0.91 at 1,000, 1,500 and 2,000 COP/km. Figure 2 shows the curve with its interval and the two regimes; the full estimation is in the online supplement. The statutory minimum hourly wage is $w^* = 4{,}367.91$ COP (2021 monthly minimum divided by 48×4.33 hours). Emission factors are 0.114 kg CO_2e/km for motorcycles and 0.167 for cars (Department for Energy Security and Net Zero and Department for Environment, Food & Rural Affairs (UK) 2023), zero for bicycles and walking; local factors by vehicle type do not exist. Table C.5 in Appendix C.1 summarises the parameters with their sources and, for the behavioural ones, the evidence grade (A: consistent across at least 15 interviewees or clear convergence of both sources; B: 5–14 interviewees or a single solid source; C: fewer than 5, or a small-n estimate with a wide interval).

5.2. Benchmark policies

We compare the three proposed policies against four benchmarks designed to isolate the value of the equity history, of the exact constraint and of the pay lever. All benchmarks share the round,

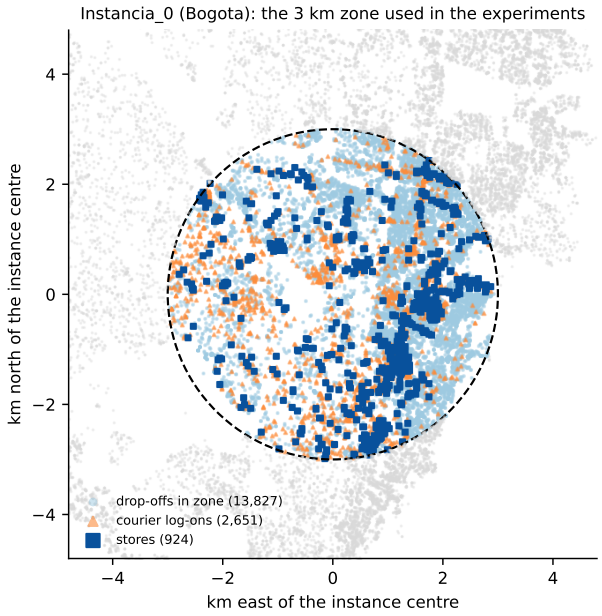


Figure 1: A 3-km dispatching zone in Bogotá

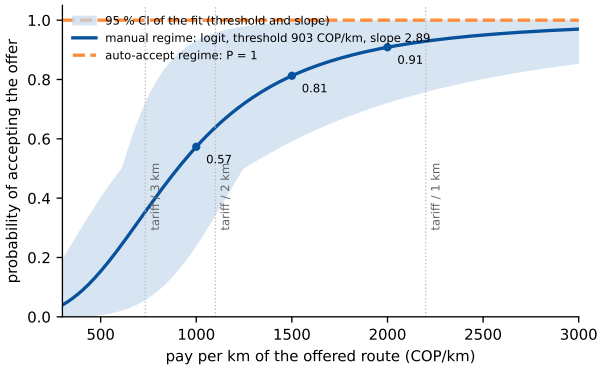


Figure 2: Acceptance probability as a function of pay per kilometre

the candidate routes, the acceptance model and the tariff of Section 4.1.

- **Greedy**: Courier-centric. The free couriers are visited in increasing order of the orders they have delivered so far, and each takes the best feasible route (a bundle when one is feasible, otherwise the nearest order) at its nearest store with pending orders. This is the platform's current practice, in which the free couriers pick work in turn, and the reference for all paired comparisons; it reads one number of $\mathcal{H}_t$, the order count, as a priority rule, and its end-of-day Gini coefficient is the historical anchor for α.
- **ILP**: A per-round integer programme over K_t that maximises the number of orders covered, with a secondary term for travel time. It is the exact corporate dispatcher and the model that **MIP-Fair** extends; comparing the two isolates the value of the Gini constraint.
- **BRKGA**: The coverage BRKGA of Section 4 with $w = 0$ and the tariff pay. Comparing it with **BRKGA-Fair** isolates the deficit term, and with **Pay-BRKGA** the pay gene.
- **Hierarchical**: A per-round lexicographic policy that first maximises coverage, then minimises lateness, then balances earnings among the couriers of the round, then minimises cost. It is fair within each window and ignores $\mathcal{H}_t$; it is the empirical counterpart of the example in Appendix B.1, and comparing it with **BRKGA-Fair** and **MIP-Fair** isolates the value of carrying the history in the state.

5.3. Experimental design

Table C.7 in Appendix C.1 lists the factors and levels. Eighteen policy configurations (four benchmarks, four weights of **BRKGA-Fair**, six thresholds of **MIP-Fair** and four pay weights of

Pay-BRKGA) are crossed with the two acceptance regimes, the 24 instances and ten seeds. Seeds are common random numbers: every comparison between policies is paired by (instance, seed), and the seed drives the acceptance draws, the decision delays and the BRKGA. The α grid is centred on the inequality band of Section 3.1 and was confirmed on the pilot, where **Greedy** gives a mean end-of-day Gini coefficient of 0.409 over the six cities.

The study consists of four experiments. **X0**, the pilot, runs one instance per city for the seven policies at default settings under the manual regime with one seed (42 runs) to measure run times and the corporate Gini coefficient. **X1**, the main experiment, runs the 18 configurations for both regimes on the 24 instances with ten seeds (8,640 runs) and produces the frontiers, the cost of the guarantee, the compliance and the regime loss. **X2**, the threshold experiment, re-runs five selected configurations (**Greedy**, **Hierarchical**, **BRKGA-Fair** with $w = 40$ and $k_d = 25$, **MIP-Fair** with $\alpha = 0.30$ and $k_d = 25$, **Pay-BRKGA** with $\lambda_{\text{pay}} = 0.5$) at the bounds of the threshold's confidence interval under the manual regime, the only one in which the threshold is read, with five seeds (1,200 runs). **X3** evaluates, with five seeds, the second-generation variants of Section 6.1 (exposure basis, deficit-aware prefilter with $k_d \in \{8, 25\}$ and coverage slack; eight configurations under both regimes, 1,920 runs) and a supplement of eleven ablations (decoder score, population size, scope and basis of the Gini constraint, equity-first priority, prefilter $k = 25$, coverage slack, and $k_d = 50$; manual regime except $k_d = 50$, 1,453 runs completed; Appendix C.4). In total 13,255 runs and 1,239 CPU-hours: a run took 2.9 minutes on average in X1, 7.1 in the second generation of X3 and in X2, and 18.7 in the ablations.

5.4. Metrics and analysis

For every run we record the two components of the cost of Eq. (5) separately, coverage (fraction of orders delivered) and mean and total lateness, the number of late orders, click-to-door mean and 90th percentile, the end-of-day F_{Gini}, F_{MaxMin}, F_{Orders} and F_{Deficit}, total payment, the number of active couriers, the mean bundle size, the mean and maximum solve time per round, and the wall-clock time; emissions follow from the per-courier distances by vehicle. The reported Gini coefficient is one and the same for all policies, Eq. (7) over the eligible couriers, computed ex post from the per-courier table of every run; the active-courier statistic $F^{\text{act}}_{\text{Gini}}$ is reported beside it, and the scope used inside a policy affects what it optimises, not what is measured. To compare the policies, we compute for each policy π and instance i the average $\bar{y}^{\pi}_{i}$ of a metric over the ten seeds and report the mean over the 24 instances together with its standard error, $\text{SE} = \text{SD}/\sqrt{N}$ with $N = 24$ and SD the sample standard deviation across instances. The cost of the guarantee is the paired difference $\bar{y}^{\pi}_{i} - \bar{y}^{\mathbf{Greedy}}_{i}$ with a 95% bootstrap interval over the 24 instances (instance as block, seeds averaged within), complemented by a paired Wilcoxon test. Compliance is the fraction of (instance, seed) pairs with $F_{\text{Gini}} \leq \alpha$ by policy and α, computed ex post, so the grid of α can be extended without new runs. The regime loss is the paired difference manual minus auto-accept. Frontiers are the mean over (instance, seed) of $(F_{\text{Gini}}, \text{coverage})$ and $(F_{\text{Gini}}, \text{lateness})$ at each level

of the swept parameter, with the non-dominated envelope and the knee selected by the maximum distance from the chord between the extremes, applied identically to the three policies.

5.5. Implementation details

The parameters of the policies were tuned in preliminary experiments. The BRKGA uses a population of 10 and 10 generations per round, elite and mutant fractions of 20%, elite bias 0.7 and $K = 3$ couriers per order in the decoder; a population of 20 with 20 generations is an ablation. The candidate prefilter keeps the $k = 25$ nearest available couriers per route for all policies except **MIP-Fair**, which uses $k = 8$ for budget reasons: with $k = 25$ a two-hour window of Bogotá did not finish within seven minutes, while with $k = 8$ it takes 4.5 minutes and covers 0.443 against 0.437 for **Greedy**; $k = 25$ is an ablation. **Pay-BRKGA** uses $\lambda_{\text{def}} = 0.5$. Mixed-integer programmes are solved with a limit of 10 seconds per round, and each run has a timeout of 90 minutes in X1, 150 in X2 and in the second generation of X3, and 240 in the ablations of X3. All experiments were conducted on a computer equipped with an Intel Core i9-14900K processor with turbo disabled, running Windows 10, with 12 parallel processes. The implementation was carried out in Python 3.10, and Gurobi 11.0 was used for the **ILP**, **MIP-Fair** and **Hierarchical** policies.

6. Computational study

In this section, we present our computational study. We first compare the policies with respect to coverage, punctuality and the end-of-day Gini coefficient, and report whether they comply with the guarantee. Then, we perform an in-depth analysis of the equity history: how the Gini coefficient evolves within the day under the per-window policy and under the policies that read $\mathcal{H}_t$, and how the three policies change who receives which route and what it pays. Finally, we examine how policy performance changes across operational settings: the acceptance regime, the behavioural threshold, the tolerated inequality and the fleet composition of the six cities.

6.1. Solution quality and compliance

To assess the solution quality of the policies, we use the **Greedy** dispatcher as the base benchmark, since it reproduces the platform's practice, and report for each policy its best configuration per regime (the configuration that satisfies the band criterion $\alpha = 0.30$ on most instances at the least coverage cost, or, where none does, the one with the lowest Gini coefficient). Table 2 reports, by regime, the mean over the 24 instances with its standard error; the best value per column and regime is in bold.

Three facts organise the table. First, under the auto-accept regime coverage is identical for the 18 configurations, 0.786 ± 0.013: the orders that are not delivered (21%) cannot be delivered on time by any courier of the zone, so the fleet is not binding and the policies differ only in who delivers, when, and at what pay. Under the manual regime rejection opens a spread in coverage,

Policy	Coverage mean	Coverage SE	End-of-day Gini F_{Gini} mean	End-of-day Gini F_{Gini} SE	Active eligible	Gini $F^{\text{act}}_{\text{Gini}}$	Lateness (min)	Min.-wage deficit F_{Deficit}	Payment (% of tariff)	CPU time (min/run)
Manual regime										
Greedy	0.758	0.013	**0.301**	0.008	**0.97**	0.509	19.9	0.981	100	2.2
ILP	0.738	0.013	0.521	0.015	0.73	**0.414**	19.4	**0.993**	100	**1.9**
Hierarchical	0.769	0.013	0.527	0.011	0.77	0.530	**18.1**	0.981	100	3.2
BRKGA	0.770	0.013	0.565	0.014	0.70	0.488	18.5	0.984	100	2.3
BRKGA-Fair ($w = 20$)	0.770	0.013	0.537	0.014	0.75	0.500	18.6	0.985	100	2.4
MIP-Fair ($\alpha = 0.30$)	0.764	0.013	0.544	0.013	0.72	0.460	19.0	0.988	100	4.5
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.753	0.013	0.546	0.013	0.71	0.457	18.8	0.991	**88**	2.8
Auto-accept regime										
Greedy	0.786	0.013	**0.296**	0.008	**0.97**	0.586	19.0	0.973	100	1.8
ILP	0.786	0.013	0.501	0.017	0.75	**0.395**	18.8	**0.993**	100	**1.4**
Hierarchical	0.786	0.013	0.515	0.010	0.79	0.570	**17.9**	0.977	100	2.8
BRKGA	0.786	0.013	0.566	0.014	0.70	0.509	18.1	0.982	100	2.0
BRKGA-Fair ($w = 20$)	0.786	0.013	0.539	0.014	0.75	0.522	18.2	0.982	100	2.1
MIP-Fair ($\alpha = 0.30$)	0.786	0.013	0.534	0.015	0.73	0.450	18.8	0.988	100	3.2
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.786	0.013	0.535	0.013	0.73	0.456	18.3	0.992	**80**	2.2

Table 2: Comparison of the policies by acceptance regime: mean over the 24 instances (seeds averaged within each instance) and standard error. F_{Gini} is the Gini coefficient of hourly earnings over the eligible couriers (connected ≥ 1 h, idle ones at zero), "active eligible" the share of eligible couriers that delivered at least one order, and $F^{\text{act}}_{\text{Gini}}$ the same Gini coefficient restricted to them. Payment is relative to the flat tariff of the orders delivered.

from 0.738 (**ILP**) to 0.770 (**BRKGA** and **BRKGA-Fair**). Second, **Greedy** is dominated on the service side: every other policy except **ILP** delivers at least as much with less lateness (-0.9 to -1.8 minutes, paired intervals away from zero). Third, and this is the central result of the paper, **Greedy** is the most equitable policy by a wide margin on the statistic of Eq. (7): 0.301 under the manual regime and 0.296 under auto-accept, against 0.501–0.566 for every other policy, the three proposed ones included. The paired differences with respect to **Greedy** are between +0.205 and +0.270, every 95% interval at least 0.17 from zero, and the ranking holds in each of the six cities.

The mechanism is who gets work. **Greedy** is courier-centric: every free courier takes its best route, the least-served courier first, so 97% of the eligible couriers deliver at least one order during the day. The exact **ILP**, **MIP-Fair**, the BRKGA family and **Hierarchical** are order-centric: each round assigns the orders to the best-placed couriers, the same couriers tend to be best placed round after round, and between 21% and 31% of the eligible couriers end the day without a single delivery and with $w_c = 0$. The first quartile of hourly earnings among eligible couriers is 481 COP under **Greedy** and between 84 and 167 COP under the other policies. Table C.10 in Appendix C.3 shows how this inverts the statistic that the fairness literature, and the first version of this study, computed over the active couriers: $F^{\text{act}}_{\text{Gini}}$ is lowest for **ILP** (0.414), then **Pay-BRKGA** and **MIP-Fair** (0.457, 0.460), and highest for **Greedy** (0.509). Two manoeuvres produce the inversion. Idle couriers leave the active population, so concentrating work on fewer couriers lowers $F^{\text{act}}_{\text{Gini}}$ while raising F_{Gini}; and the 7% of couriers who connect for less than one hour, to whom **Greedy** dispatches because they are free and nearby, enter $F^{\text{act}}_{\text{Gini}}$ with hourly earnings of up to 110,000 COP (one delivery of 2,200 COP in a one-minute connection) and raise it, whereas Eq. (7) excludes them. Restricted to the eligible couriers that were active, **Greedy** is still the most equitable (0.28 against 0.35–0.39), so the first manoeuvre is the larger one.

The equity history did not prevent this because of what the three policies read in it. The

deficit of **BRKGA-Fair** and **Pay-BRKGA**, and the projected w_c of **MIP-Fair**, are computed per *active* hour, minutes in motion or serving, which is a productivity rate: a courier with one ten-minute delivery and a courier with twenty have nearly the same value, and only a courier who has never delivered has a deficit. The signal therefore separates couriers with and without work but not couriers with little and much, and it competes with coverage, which has lexicographic priority in **MIP-Fair** and is the fitness of the BRKGA population. The deficit bias does act in the right direction: **BRKGA-Fair** raises the active share of eligible couriers from 0.70 (**BRKGA**) to 0.75 and lowers F_{Gini} from 0.565 to 0.534–0.547 as w grows from 5 to 40, in both regimes, and **MIP-Fair** activates more couriers than **BRKGA**; but the effect is a fraction of the gap to **Greedy**. The α of **MIP-Fair** has no measurable effect (0.544–0.545 under the manual regime, including $\alpha = 1.00$, which switches the constraint off), because with coverage first and $k = 8$ candidates the coverage-optimal assignments of a round leave the Gini block almost nothing to choose. The second-generation variants that correct this are evaluated after Table C.10. The minimum-wage deficit is between 0.973 and 0.993 under every policy, consistent with the 92% of surveyed couriers below the statutory minimum. On CPU time, **ILP** is the cheapest policy and **MIP-Fair** the most expensive (4.5 minutes per day on average, 22 at most on a Bogotá instance); the three proposed policies cost between 1.1 and 2.1 times **Greedy**.

Second-generation variants. Two changes follow from this diagnosis. First, the history $\mathcal{H}_t$ that the three policies read is the *exposure* basis of Eq. (7), earnings per hour connected so far over the eligible couriers, so that an idle eligible courier carries the largest deficit. Second, a deficit-aware prefilter adds to the candidate set of each route, after its k nearest free couriers, the k_d free couriers with the largest deficit within the dispatching radius, so that the courier who should receive the route is at least offered to the solver. Table 3 reports the variants on the five seeds shared with X1, paired with their first-generation parents on the same (instance, seed) pairs. The basis alone is almost inert: it moves F_{Gini} by 0.004 (**MIP-Fair**) to 0.025 (**BRKGA-Fair**) and closes 2–12% of the gap to **Greedy**, because an idle courier who is not a candidate cannot be chosen however large its deficit. The prefilter is the lever, and its effect grows with k_d: with $k_d = 8$ the gap closes by 25% (**BRKGA-Fair**) and 33–46% (**MIP-Fair**); with $k_d = 25$ by 45–54% (**BRKGA-Fair**, $w = 40$) and 47–57% (**MIP-Fair**), to $F_{\text{Gini}} = 0.435$–0.437 under the manual regime and 0.401–0.416 under auto-accept, with 85–91% of the eligible couriers active and a first quartile of hourly earnings of 282–343 COP/h, two to three times that of the parents. The 95% interval of the paired difference with the parent excludes zero in every case, no (instance, seed) pair is worse than its parent at $k_d = 25$, and the per-seed means differ by at most 0.008. The service cost is nil: coverage changes by -0.001 to $+0.003$ with respect to the parent (the **MIP-Fair** variants gain slightly, because the extra candidates also serve coverage), lateness rises by 0.1–0.3 minutes, and the regime loss is unchanged. The coverage slack ε of **MIP-Fair** does nothing ($\varepsilon = 0.05$ against 0: F_{Gini} within 0.002 and coverage within 0.001), which confirms that what bound the Gini block

was the candidate set and not the lexicographic priority. **Pay-BRKGA** responds least to the prefilter (9–12% of the gap) because its deficit term is half of a fitness dominated by pay. The price of k_d is CPU time: **BRKGA-Fair** with $k_d = 25$ costs 4.3–4.9 minutes per day on average and 25 at most, twice its parent; **MIP-Fair** with $k_d = 25$ costs 11–24 minutes on average and up to 110 on a Bogotá day under the manual regime, five times its parent, for the same end-of-day Gini, so the heuristic is the practical choice at this scale. Half of the gap to **Greedy** remains, and the ablation with $k_d = 50$ (Table C.11 in Appendix C.4) shows that for **BRKGA-Fair** it is a matter of candidates, since F_{Gini} keeps falling to 0.404 under the manual regime and 0.352 under auto-accept at unchanged coverage and 1.3 times the CPU time, whereas for **MIP-Fair** the curve flattens (0.412 and 0.398) while the CPU time grows to 51 minutes per day: for the exact round model the one-route-per-courier structure and the solver limit bind before the candidate set does.

Policy	Coverage mean	Coverage SE	End-of-day Gini F_{Gini} mean	End-of-day Gini F_{Gini} SE	Active eligible	Gini $F^{\text{act}}_{\text{Gini}}$	Lateness (min)	Min.-wage deficit F_{Deficit}	Payment (% of tariff)	CPU time (min/run)
Manual regime										
Greedy	0.758	0.013	**0.301**	0.008	**0.97**	0.507	19.9	0.981	100	**2.2**
BRKGA-Fair ($w = 40$)	0.770	0.013	0.534	0.015	0.75	0.504	18.6	0.986	100	2.4
BRKGA-Fair ($w = 20$, exposure, $k_d = 8$)	0.770	0.013	0.483	0.016	0.80	0.432	18.7	0.993	100	4.6
BRKGA-Fair ($w = 40$, exposure, $k_d = 25$)	0.769	0.013	0.437	0.016	0.85	**0.398**	18.8	**0.995**	100	4.9
MIP-Fair ($\alpha = 0.30$)	0.763	0.013	0.544	0.013	0.72	0.458	19.0	0.988	100	4.6
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 8$, $\varepsilon = 0.00$)	0.769	0.013	0.465	0.012	0.86	0.473	19.1	0.990	100	11.3
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 25$, $\varepsilon = 0.00$)	0.767	0.013	0.435	0.014	0.88	0.466	19.3	0.990	100	23.5
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.753	0.013	0.547	0.013	0.71	0.456	18.8	0.991	88	2.8
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$, exposure, $k_d = 8$)	0.753	0.013	0.526	0.014	0.74	0.445	18.9	0.992	88	5.2
Auto-accept regime										
Greedy	0.786	0.013	**0.296**	0.008	**0.97**	0.586	19.0	0.973	100	**1.8**
BRKGA-Fair ($w = 40$)	0.786	0.013	0.536	0.015	0.76	0.512	18.2	0.983	100	2.0
BRKGA-Fair ($w = 20$, exposure, $k_d = 8$)	0.786	0.013	0.482	0.016	0.80	0.442	18.2	0.992	100	4.1
BRKGA-Fair ($w = 40$, exposure, $k_d = 25$)	0.786	0.013	0.416	0.016	0.87	**0.390**	18.3	**0.995**	100	4.3
MIP-Fair ($\alpha = 0.30$)	0.786	0.013	0.534	0.015	0.73	0.448	18.8	0.988	100	3.2
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 8$, $\varepsilon = 0.00$)	0.786	0.013	0.426	0.011	0.90	0.458	18.8	0.991	100	6.9
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 25$, $\varepsilon = 0.00$)	0.786	0.013	0.401	0.011	0.91	0.456	18.8	0.990	100	11.1
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.786	0.013	0.534	0.013	0.73	0.454	18.3	0.992	80	2.2
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$, exposure, $k_d = 8$)	0.786	0.013	0.507	0.013	0.77	0.440	18.3	0.994	80	4.2

Table 3: Second-generation variants (exposure basis over the eligible couriers; deficit-aware prefilter with k_d extra candidates; coverage slack ε in **MIP-Fair**) against their first-generation parents and **Greedy**, on the five seeds of X3 (1,920 runs) and the same seeds of X1: mean over the 24 instances and standard error, columns as in Table 2. Best value per column and regime in bold.

Table C.9 in Appendix C.3 reports the compliance with the guarantee, the fraction of (instance, seed) pairs with $F_{\text{Gini}} \leq \alpha$, by policy and α. **Greedy** meets $\alpha = 0.30$, the inside-the-ladder anchor of Section 3.1, on half of the days, 0.35 on nine days in ten and 0.40 on every day under auto-accept. No other configuration meets $\alpha \leq 0.40$ on any day under either regime; at 0.45 the best of them (**ILP**) reaches 18% of the days under the manual regime and 46% under auto-accept, and at 0.50 the proposed policies reach 35–50%. The lower half of the α grid is therefore within reach of a dispatcher that serves the whole day, but only of the one that spreads the work; the parameter of each proposed policy moves compliance by a few points at most (w in **BRKGA-Fair**: 38–46% at $\alpha = 0.50$; α in **MIP-Fair**: 39–40%; λ_{pay} in **Pay-BRKGA**: 35–46%, falling as pay falls). The second generation changes the lower half of the grid (last block of the table): with $k_d = 25$, **BRKGA-Fair** and **MIP-Fair** meet $\alpha = 0.40$ on 40–49% of the days under the manual regime and 56–63% under auto-accept, 0.45 on 61–62% and 67–83%, and 0.50 on 72–82% and 81–96%,

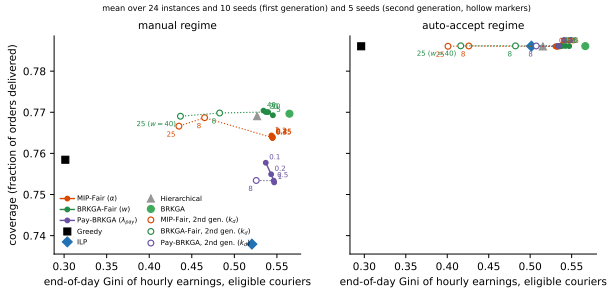


Figure 3: Coverage against end-of-day Gini by policy; hollow markers and dotted paths: second-generation variants along k_d (X3, five seeds)

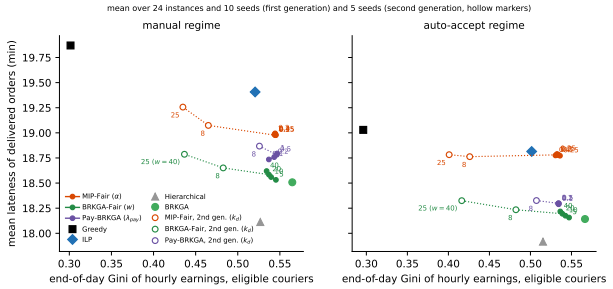


Figure 4: Lateness against end-of-day Gini by policy; hollow markers and dotted paths: second-generation variants along k_d (X3, five seeds)

and still on none at 0.30. We therefore take $\alpha^* = 0.45$ as the operating point of the threshold experiment of Section 6.4: the smallest value of the grid that a second-generation policy meets on most days under both regimes at no coverage cost, inside the interval of Table C.13.

Figures 3 and 4 show the frontiers traced by the three policies in their own parameter: α for **MIP-Fair**, w for **BRKGA-Fair** and λ_{pay} for **Pay-BRKGA**. The x-axis is the end-of-day Gini coefficient of Eq. (7), the y-axis coverage (Figure 3) and mean lateness (Figure 4), each point is a configuration averaged over the 24 instances and the seeds available, and the benchmarks appear as single points. The three frontiers occupy a narrow band on the right of the figures, between 0.53 and 0.57, and **Greedy** stands alone on the left at 0.30 with less coverage and more lateness than any of them: no configuration of the three policies trades service for equity along its own parameter at a scale comparable to the gap between the two groups. **BRKGA-Fair** is the only sweep with a monotone direction, moving 0.03 to the left as w grows at constant coverage; **MIP-Fair** collapses to a point in α; **Pay-BRKGA** moves along coverage (manual) and lateness rather than along the Gini axis as λ_{pay} grows. Under auto-accept all configurations share the same coverage and the coverage frontier degenerates into the Gini axis. The second-generation variants (hollow markers, five seeds) populate the region between the two groups along k_d rather than along the original parameters: from the parent at 0.53–0.54 to 0.47–0.48 at $k_d = 8$ and 0.44 at $k_d = 25$ under the manual regime (0.40–0.42 under auto-accept), at constant coverage and with lateness within 0.3 minutes of the parent. The gap that the first-generation parameters could not close is halved by a change in the candidate set, not in the objective; **Pay-BRKGA** moves least (0.02).

6.2. The equity history within the day

One of our main modelling contributions is the equity history in the state. In this section, we analyse how the Gini coefficient evolves within the day under the policies that read $\mathcal{H}_t$ and under the policy that does not. Figure 5 shows the trajectory of the Gini coefficient of hourly earnings so far, computed every round over the eligible couriers connected up to that round, idle ones at zero, for **Greedy**, **Hierarchical**, **MIP-Fair** ($\alpha = 0.30$) and the second-generation **BRKGA-Fair** of

Section 6.4 ($w = 40$, deficit-aware prefilter with $k_d = 25$) on one Bogotá day (instance 0, manual regime, seed 1: 10,215 orders, 2,650 couriers in the zone, of whom 2,560 are eligible). The x-axis is the time of day, the y-axis the Gini coefficient, recomputed every ten minutes from the delivery log; the dashed line marks $\alpha = 0.40$, the value that **Greedy** meets on every day under auto-accept. The left panel is the statistic of the paper; the right panel is the same coefficient over the couriers that have delivered at least once so far, the population on which the first-generation policies read their history. The three trajectories start near 1: at 08:00, 525 eligible couriers are connected and about forty have delivered, so almost everyone is at zero, and the coefficient falls as orders arrive and the fleet is put to work. The policies separate between 10:00 and 16:00, the hours in which most of the fleet connects. **Greedy** puts the new couriers to work as they arrive (at 14:00, 2,137 of the 2,311 eligible couriers connected have delivered) and its trajectory falls from 0.79 at 10:00 to 0.41 at 14:00, crosses $\alpha = 0.40$ at 14:12 and settles at 0.23 from 16:00 on. **Hierarchical** and **MIP-Fair** keep assigning to the couriers already in motion (1,414 and 1,480 active at 14:00), so their trajectories fall more slowly, 0.59 and 0.57 at 14:00, and never reach α: they end the day at 0.48 and 0.47 with 16% and 15% of the eligible couriers idle. The right panel shows why neither policy reacted. On the active couriers, **Hierarchical** is at 0.36–0.39 all afternoon and **MIP-Fair** at 0.33–0.36, below α and below **Greedy** (0.34–0.39): the per-window statistic that **Hierarchical** controls, and the projected statistic that **MIP-Fair** constrains, both report compliance while the end-of-day statistic of Eq. (9) is 0.2 above the guarantee, because the couriers who make the difference are not in their population. The second-generation **BRKGA-Fair**, which admits the idle eligible couriers with the largest deficit as candidates in every round, follows the order-centric trajectory until noon and then bends towards **Greedy**: 0.51 at 14:00, below α at 15:48, and 0.36 at the end of the day with 97% of the eligible couriers active and coverage 0.780, the highest of the four. The comparison answers the first research question: the per-window statistic and the end-of-day statistic move in different directions, because the first is computed on the couriers of the round and the second on the couriers of the day, and the end-of-day one is decided by the hours in which the fleet grows.

6.3. Decision making

We next analyse how the three policies change the dispatching decision compared to **Greedy**: who receives which route, and what it pays. Figure C.6 in Appendix C.3 compares the distribution of end-of-day hourly earnings over the eligible couriers under **Greedy** and under each policy at its best configuration, pooled over the 24 instances of one seed (23,944 couriers, manual regime); the x-axis lists the policies, the y-axis the earnings per connected hour, and the share of couriers who end the day idle is printed under each violin. The statutory minimum $w^* = 4{,}368$ COP/h lies above the 99th percentile of every distribution, which is the content of $F_{\text{Deficit}} \geq 0.97$. The policies do not compress the distribution of **Greedy**; they stretch it at both ends. **Greedy** leaves 3% of the eligible couriers idle and its interquartile range is 519–876 COP/h around a median of 685; the six

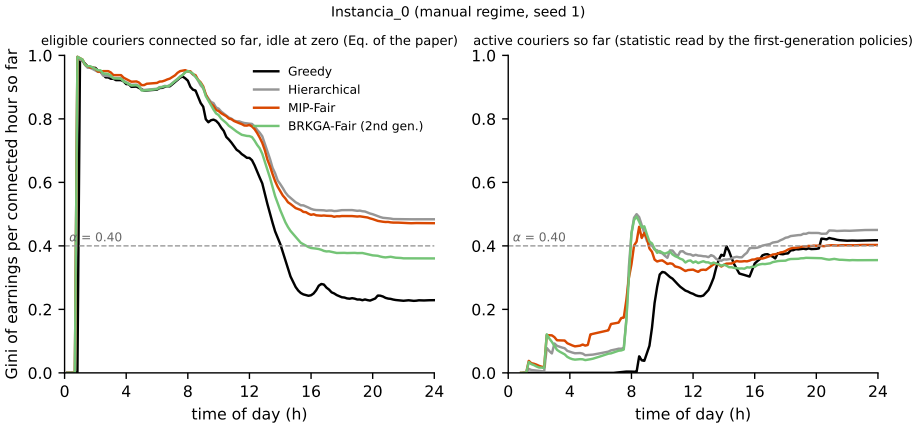


Figure 5: Intraday trajectory of the Gini coefficient of earnings per connected hour on one Bogotá day (instance 0, manual regime, seed 1), recomputed every ten minutes: over the eligible couriers connected so far with idle ones at zero (left, the statistic of Eq. (7)) and over the couriers with at least one delivery so far (right). Dashed line: $\alpha = 0.40$.

other policies leave 19–25% idle, their first quartile is between 0 (**BRKGA**) and 205 COP/h, their median is similar (547–621) and their third quartile is higher (1,066–1,206 COP/h): the orders that **Greedy** spreads over the couriers who happen to be free go, under an order-centric policy, to the couriers who are best placed, who are then best placed again. The reduction of the Gini coefficient of Eq. (7) under **Greedy** therefore comes from below (fewer idle couriers) and from above (fewer high earners) at once, at the same total pay: the mean is 801 COP/h under **Greedy** and 708–796 under the policies that pay the tariff. **Pay-BRKGA** shifts the whole distribution down (mean 689, median 547) because it pays 12% less, and still leaves 23% idle. No courier crosses w^* under **Pay-BRKGA** either. Bundling plays no role in these differences: the mean bundle size is 1.03 orders per route under **Greedy** and the BRKGA-type policies, 1.01 under **MIP-Fair** and 1.00 under the **ILP** and **Hierarchical** policies. For **Pay-BRKGA** the pay goes where acceptance is at stake. The total payment is 88% of the tariff under the manual regime and 80% under auto-accept (Table 2), while the median pay per kilometre received by an active courier is 929 COP/km against 816 under **Greedy** in the manual regime (789 against 669 under auto-accept): the multiplier raises the pay of the long routes that would otherwise be rejected and lowers that of the short ones that would not, so the bill falls while the pay per kilometre of what is accepted stays above the threshold. In the auto-accept regime, where nothing is rejected, the same rule only lowers the bill and leaves F_{Deficit} unchanged.

6.4. Sensitivity analysis across operational settings

Finally, we study how different operational conditions affect the performance and relative advantages of the policies. We vary one factor at a time around the base case (manual regime, $\theta = 903$ COP/km, the α grid, circuity 1.3) and report, for the benchmark dispatchers and the proposed

policies at their selected configurations, the relative difference over the base-case value of the policy, following the one-factor-at-a-time design. Table 4 reports the results for coverage; the regime and α columns come from X1, the θ columns from X2, which re-runs five configurations at the ends of the 95% interval of the logit, and the circuity column from the pilot, where only **Greedy** was run at circuity 1.0.

	Relative difference over the base-case setting (in %)									
Policy\Setting	**Regime**		**Threshold** θ (COP/km)			**Tolerated inequality** α			**Circuity**	
	manual	**auto**	**615**	**903**	**1,242**	**0.20**	**0.30**	**0.40**	**1.0**	**1.3**
Greedy	0.0	+3.6	+3.2	0.0	−9.0	–	–	–	+6.0	0.0
Hierarchical	0.0	+2.2	+2.0	0.0	−5.5	–	–	–	–	0.0
MIP-Fair ($\alpha = 0.30$)	0.0	+2.9	–	0.0	–	0.0	0.0	0.0	–	0.0
BRKGA-Fair ($w = 40$, $k_d = 25$)	0.0	+2.2	+2.0	0.0	−5.7	–	–	–	–	0.0
MIP-Fair ($\alpha = 0.30$, $k_d = 25$)	0.0	+2.5	+2.3	0.0	−7.0	–	–	–	–	0.0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.0	+4.3	+3.6	0.0	−7.3	–	–	–	–	0.0

Table 4: Sensitivity analysis across operational settings: relative difference of coverage over the base case (manual regime, $\theta = 903$, $\alpha = 0.30$, circuity 1.3). Regime and α from X1 (ten seeds); θ from X2 (five seeds, the five configurations of the threshold experiment); circuity from the pilot (**Greedy**, one instance per city). The companion for the Gini coefficient is Table C.12 in Appendix C.5

Acceptance regime. Figure C.7 in Appendix C.5 shows the regime loss, the paired difference manual minus auto-accept by instance, for the seven configurations of Table 2. In coverage the loss is between 1.6 points (**BRKGA**, **BRKGA-Fair**) and 4.8 points (**ILP**), with 2.8 for **Greedy**, 2.2 for **MIP-Fair** and 3.3 for **Pay-BRKGA**; every interval is away from zero, and the loss of **ILP** alone exceeds the whole spread of coverage between policies within the manual regime (3.2 points). The policies that re-optimise the round after a rejection lose least. **Pay-BRKGA** loses 1.7 points more than **BRKGA**: its pay cut lowers the acceptance probability of the long routes on which the multiplier would have to rise, so in the manual regime the pay lever is paid for in coverage, whereas under auto-accept the same lever cuts the bill by 20% at no coverage cost. The regime moves the end-of-day Gini coefficient by at most 0.02 (+0.019 for **ILP**, −0.002 for the BRKGA family), an order of magnitude less than the gap between **Greedy** and the other policies, so the ranking of Table 2 is the same under both regimes. With a flat tariff and a threshold of 903 COP/km, a 4-km singleton is accepted with probability 0.19, so coverage under manual acceptance is bounded by behaviour rather than by dispatching, and auto-accept removes that bound by moving the courier's decision to connection time: the regime matters more than the policy for coverage, and the policy matters more than the regime for equity.

Behavioural threshold. The threshold θ is a grade-C parameter estimated from 32 decisions, so X2 re-runs the five configurations of Section 5.3 at the ends of its 95% interval, $\theta = 615$ and 1,242 COP/km, paired with the same seeds at $\theta = 903$ (three **MIP-Fair** runs on the largest Bogotá day exceeded the timeout at $\theta = 1{,}242$). With a flat tariff of 2,200 COP per order, a single order is accepted with probability one half at 3.6 km when $\theta = 615$, at 2.4 km when $\theta = 903$ and at 1.8 km

when $\theta = 1{,}242$, so the response of coverage is asymmetric: the lower threshold adds 1.5–2.7 points (+2.0 to +3.6%) and the upper one removes 4.2–6.9 points (−5.5 to −9.0%). The policies that re-optimise the round after a rejection lose least (**Hierarchical** −5.5%, **BRKGA-Fair** −5.7%) and **Greedy** loses most (−9.0%), so the coverage ranking changes at the upper end of the interval: **Greedy**, fourth of five at $\theta = 903$, is last at 1,242, 3.7 points below **Hierarchical**, and **Pay-BRKGA**, last at 903, overtakes it because its multiplier keeps the long routes above the threshold. At $\theta = 615$ the five configurations cover within half a point of each other. The end-of-day Gini coefficient is less sensitive than coverage, as expected from a statistic normalised by connected time: it moves by −0.7 to −2.8% at 615 and by +2.2 to +3.8% at 1,242, every paired interval away from zero but no difference larger than 0.017, and the fairness ranking is the same at the three thresholds (**Greedy** 0.30–0.31, the two second-generation policies 0.42–0.45, **Hierarchical** 0.52–0.54, **Pay-BRKGA** 0.54–0.56). Compliance with $\alpha = 0.45$ is 63%, 62% and 52% of the days for **BRKGA-Fair** and 73%, 61% and 50% for **MIP-Fair** at 615, 903 and 1,242: the recommendation of Table C.13 holds inside the interval, with ten points of compliance lost at its upper end.

Tolerated inequality and the choice of α. Three complementary criteria are used to choose α. Criterion A (historical anchor) sets α equal to the end-of-day Gini coefficient observed under **Greedy**, so the fair policy must be no less equal than the status quo; on the ten seeds of X1 this is 0.30 (0.27–0.32 by city). Criterion B (inequality bands) sets α at the upper bound of a conventional severity band, a Gini coefficient below about 0.30 read as low and above about 0.40 as high; the survey anchors are 0.26 inside the bonus ladder and 0.39 outside it, so $\alpha = 0.30$. Criterion C (Pareto knee) sweeps α and selects the point of maximum distance from the chord between the extremes of the coverage–Gini frontier. Table C.13 in Appendix C.5 reports α under the three criteria, the policy that satisfies each at the least coverage cost, and that cost. Criteria A and B coincide at 0.30 on these data, because the corporate dispatcher is itself inside the low-inequality band, and the only policy that meets it is **Greedy**, on half of the days. Criterion C does not apply: coverage does not fall along either sweep, so there is no knee to find. The interval is therefore $[0.30, 0.45]$: 0.30 is the status quo, which the order-centric policies do not reach on any day; 0.40, the upper bound of the band, is met on 40–63% of the days by the second generation at no coverage cost; and 0.45 is the smallest value met on most days under both regimes. We recommend operating at $\alpha^* = 0.45$ with **BRKGA-Fair** ($w = 40$, $k_d = 25$), which costs 4–5 minutes of CPU per day against 11–24 for **MIP-Fair** at the same compliance, and keeping the courier-centric rule as the reference for $\alpha \leq 0.35$.

Fleet composition and emissions. Fairness in hourly earnings is neutral to vehicle type by construction; any emissions effect comes from which couriers receive distance, so the diagnostic is informative precisely where fleets are mixed. Figure C.8 in Appendix C.5 reports the diagnostic from the per-courier distances and the emission factors of Table C.5. Fleet composition sets the level: a delivered order costs 0.13–0.18 kg CO_2e in Bogotá, where bicycles and walkers carry a

quarter of the distance, and 0.25–0.36 in Pereira, where motorcycles carry nine tenths. The policy sets the rest, and in the opposite direction to equity. **Greedy** drives 2.77 km per delivered order against 2.19–2.25 for the six order-centric policies, a quarter more, because every free courier takes its best route and the orders are less often bundled; it does shift distance towards bicycles and walkers (29% of its kilometres against 16–23% for the others), but the shift offsets only about 15% of the extra distance. Per delivered order, **Greedy** emits 0.236 kg, **MIP-Fair** and **ILP** 0.187–0.189 (21% less) and the BRKGA family and **Hierarchical** 0.21–0.23; the fair policies that re-optimise the assignment with an ILP are the cleanest because the distance saved by bundling dominates the fleet shift. Equity in hourly earnings, as **Greedy** produces it, is therefore paid in kilometres as well as in lateness. Additionally, Appendix C reports the results by city and the ablations of X3: the decoder score, the scope of the Gini constraint, the earnings basis, the equity-first priority, the candidate prefilter and the BRKGA size.

7. Findings and Discussion

This section addresses the three research questions and discusses their implications.

We begin by addressing the first research question concerning whether an end-of-day guarantee requires the equity history in the state. The results show that fairness enforced within each window does not deliver fairness at the end of the day: the **Hierarchical** policy equalises the pay offered among the couriers of every round and ends the day with a Gini coefficient of 0.53 over the eligible couriers against 0.30 for the corporate dispatcher, because the per-window statistic is computed on the couriers of the round and ignores both what each has already earned and who was left out of the round. The results also show that reading $\mathcal{H}_t$ is necessary but not sufficient: the three policies that read it ended the day with the same inequality as the policies that do not, because what they read, earnings per active hour, is a productivity rate that does not distinguish a courier with one delivery from a courier with twenty, and because they read it over the couriers of the round rather than over the population of the guarantee. The content of the history must be exposure, earnings per hour connected, and its population the eligible couriers, so that the deficit of an idle courier grows with every hour it waits; this is the signal that the courier-centric dispatcher obtains for free by serving the least-served courier first. The second generation of X3 confirms the diagnosis: with exposure as the content and the deficit-aware prefilter as the way to reach the idle couriers, **BRKGA-Fair** and **MIP-Fair** close 45–57% of the gap with $k_d = 25$ and **BRKGA-Fair** 59–81% with $k_d = 50$, at no coverage cost. The cost of the history remains one vector of three numbers per courier, which every platform already records.

We next address the second research question concerning which lever satisfies the guarantee at the lowest cost. On the ten seeds of X1 the answer is the assignment rule itself: the courier-centric rule meets $\alpha = 0.40$ on every day under auto-accept and on 98% of the days under the manual regime at a cost of 1.1–1.2 points of coverage and 0.5–1.8 minutes of lateness with respect to the

order-centric policies, which meet it on none. Among the three proposed policies, **BRKGA-Fair** is the only one whose parameter moves the end-of-day statistic in the right direction at no coverage cost; the α of **MIP-Fair** is inert because coverage has priority in every round; and **Pay-BRKGA** is the only policy that moves pay, which lowers the platform's bill by 12% (manual) to 20% (auto-accept) of the tariff but does not reach the couriers who are left idle. The second generation of X3 narrows but does not close this difference: reading exposure instead of activity and offering the most-deprived free couriers to the solver halves the gap to **Greedy** ($F_{\text{Gini}} = 0.44$ manual, 0.40–0.42 auto-accept) at no coverage cost and 0.1–0.3 minutes of lateness, so a second-generation **BRKGA-Fair** meets $\alpha = 0.45$ on 62–67% of the days while keeping the point of coverage that **Greedy** gives up. Part of the remaining gap is a matter of candidates and part is structural: with $k_d = 50$ the heuristic reaches 0.40 (manual) and 0.35 (auto-accept), whereas the exact round model flattens, because an order-centric round assigns at most one route per courier and the coverage objective fixes which couriers it needs. Because coverage is flat across policies at the scale of a dispatching zone, the cost of the guarantee is paid mainly in lateness and in who receives the routes. Dispatching alone cannot lift couriers to the statutory minimum wage at the current tariff; what it can do is distribute the shortfall evenly, which is the practical content of using F_{Gini} as the constraint and F_{Deficit} as a diagnostic.

Finally, we address the third research question concerning the operational conditions under which the guarantee is affordable. The acceptance regime matters more than the policy in the manual world, where coverage is bounded by behaviour; the sanctions that moved the workforce to auto-accept are the platform's way of removing that bound, and our results quantify what the bound costs. Within the confidence interval of the behavioural threshold the fairness ranking of the policies does not change, and the coverage ranking changes only at the upper end, where the courier-centric dispatcher, which does not re-optimise after a rejection, loses most (9% of its coverage against 5.5% for the policies that do). The three criteria for α give an interval rather than a point, and the recommendation is the policy that satisfies the band criterion on most days at the least coverage cost. From a managerial perspective, the central implication is that a fairness guarantee should be stated, measured and enforced on the day, not on the window, and that the pay lever and the assignment lever are complements: the first buys acceptance where couriers reject, the second distributes work where they do not.

8. Conclusion

In this paper, we study a dispatching problem in which a platform repeatedly offers work and pay to self-employed agents who may reject it, and in which the platform is accountable for the distribution of the agents' earnings at the end of the day. We examine this structure in meal delivery, where we formulate the problem as a constrained Markov decision process with an equity history in the state, an end-of-day Gini constraint on hourly earnings over the couriers connected

for at least one hour, idle ones included, and acceptance that responds to pay per kilometre in one of two regimes. As a methodological contribution, we develop three cost function approximations that read the equity history in every round, including a block-exact linearisation of the Gini constraint, and we evaluate them against four benchmarks on 24 real operating days from six Colombian cities with field-calibrated behaviour. The computational results show that fairness enforced window by window does not deliver fairness at the end of the day, that the population over which the guarantee is defined decides what the guarantee means, and that the acceptance regime bounds what any dispatching policy can achieve in the manual world. Over the eligible couriers the courier-centric dispatcher, which serves the least-served courier first, ends the day at a Gini coefficient of 0.30 and gives work to 97% of them, while every order-centric policy, including the three proposed, ends between 0.52 and 0.57 by leaving a quarter of them idle; restricted to the couriers who delivered, the same policies appear more equitable than the corporate one. Reading exposure over the eligible couriers and offering the most-deprived couriers to the solver closes about half of that gap with 25 extra candidates per route (0.44 manual, 0.40 auto-accept) and, for **BRKGA-Fair**, 59–81% with 50 (0.40 and 0.35), at no coverage cost. From a managerial perspective, the central implication is that a fairness guarantee is a property of the day: it should be measured on hourly earnings over a population the dispatcher cannot shrink, enforced through what every courier has earned per hour connected so far, and complemented by the pay lever where couriers reject.

Beyond the meal-delivery application, our contribution concerns both the decision structure and the architecture used to address it. The equity history and the end-of-day constraint apply to any sequential assignment problem in which the agents' outcomes accumulate over a horizon and the regulator or the platform cares about their distribution at its end, such as ride-hailing, crowdsourced delivery and task platforms. Within this architecture, the fairness metric can be replaced by another inequality index, the acceptance model by another behavioural model, and the per-round problem by another cost function approximation; the block linearisation applies to any constraint in mean absolute differences in which each agent's outcome takes finitely many values per round; only the per-window problem must be reformulated for the routing decisions and constraints of the new application.

The study also has several limitations related to the decision environment. The behavioural evidence is from Bogotá, and its use in the other five cities is a transfer assumption; the five cities are robustness checks, not independent calibrations. The acceptance logit rests on 32 decisions by 11 interviewees, with wide intervals, and reflects the 2020 manual regime; we report the sensitivity over the interval and run both regimes, but the parameters are grade C. The survey reports pay in bands, and 93% of couriers report that the pay shown changes after acceptance (realised-to-shown ratio 0.60); the model responds to the pay shown, so realised earnings may be lower. The equity history is intraday whereas the platform's bonus ladder is a multi-day mechanism, and the model has no sanctions, no acceptance-rate dynamics, no cash-handling constraint (couriers report a cash cap around 70–80 thousand COP) and no heterogeneity by seniority or nationality, which

the survey shows to matter. Circuity, vehicle speeds, the 30-minute slack and the UK emission factors are declared assumptions without local sources. The 3-km zone is a dispatching zone, not the city, so cross-zone effects are not modelled. Future research could extend the history to several days, add the dynamics of sanctions and acceptance rates, and calibrate the behavioural model in a second city.

There are also methodological limitations that motivate further research. **MIP-Fair** uses a candidate prefilter of eight couriers per route for budget reasons and a 10-second limit per round; a column-generation or a rolling subset of pairs could relax both. The three policies are tuned by enumeration over a grid; learning the deficit weight or the pay multiplier as a function of the state, as in Stein et al. (2025), could reduce the cost of the guarantee further, at the price of an offline training effort. The minimum-wage constraint is not imposed because it is infeasible at the current tariff; a joint design of tariff and dispatching, in which the platform's budget is a decision, is a natural extension. Finally, the 240 public instances of Reyes et al. (2018) could be run with the policies already in the simulator to test the findings outside Colombia.

Acknowledgment

[TODO: funding, the platform that provided the data, the couriers who answered the survey and the interviews.]

References

Ahamed, T., Zou, B., Farazi, N.P., Tulabandhula, T., 2021. Deep reinforcement learning for crowdsourced urban delivery: System states characterization, heuristics-guided action choice, and rule-interposing integration. Transportation Research Part B: Methodological 152, 227–257. Source: thesis.

Altman, E., 1999. Constrained Markov Decision Processes. Chapman & Hall/CRC, Boca Raton.

Auad, R., Erera, A., Savelsbergh, M., 2023. Courier satisfaction in rapid delivery systems using dynamic operating regions. Omega 121, 102917. Source: thesis.

Bajec, P., Tuljak-Suban, D., 2022. A strategic approach for promoting sustainable crowdshipping in last-mile deliveries. Sustainability 14, 13508. Source: thesis.

Bi, H., Zhu, X., Lu, F., Huang, M., 2023. The meal delivery routing problem in e-commerce platforms under the shared logistics mode. Journal of Theoretical and Applied Electronic Commerce Research TODO volume/pages. source: thesis.

Bozanta, A., Cevik, M., Kavaklioglu, C., Kavuk, E.M., Tosun Misirli, A., Sonuc, S.B., Duranel, A., Basar, A., 2022. Courier routing and assignment for food delivery service using reinforcement learning. Computers & Industrial Engineering 164, 107871. Source: thesis.

Corbett-Davies, S., Gaebler, J.D., Nilforoshan, H., Shroff, R., Goel, S., 2023. The measure and mismeasure of fairness: A critical review of fair machine learning. Journal of Machine Learning Research 24, 1–117. Source: thesis.

Cuellar-Usaquén, D., Ulmer, M.W., Gomez, C., Álvarez-Martínez, D., 2024. Adaptive stochastic lookahead policies for dynamic multi-period purchasing and inventory routing. European Journal of Operational Research 318, 1028–1041.

Dantzig, G.B., Ramser, J.H., 1959. The truck dispatching problem. Management Science 6, 80–91.

Department for Energy Security and Net Zero and Department for Environment, Food & Rural Affairs (UK), 2023. Greenhouse gas reporting: conversion factors 2023. Technical Report. UK Government. Emission factors: average motorbike 0.114 and average car 0.167 kg CO_2e/km (decision D6). TODO confirm exact figures and URL.

Giraldo-Herrera, D., Álvarez, D., 2024. A GRASP algorithm for the meal delivery routing problem, in: Metaheuristics International Conference (MIC 2024), Lecture Notes in Computer Science 14754, Springer. pp. 306–320. TODO confirm co-author first name and LNCS volume. source: thesis.

Giraldo-Herrera, D., TODO co-authors, 2026. TODO title: BRKGA with rolling horizon for the meal delivery routing problem (BRKGA-Coverage / BRKGA-Travel). Metaheuristics International Conference (MIC 2026). TODO: status (submitted/accepted), exact title. Source of the BRKGA-Coverage baseline.

Gonçalves, J.F., Resende, M.G.C., 2011. Biased random-key genetic algorithms for combinatorial optimization. Journal of Heuristics 17, 487–525.

Gupta, A., Yadav, R., Nair, A., Chakraborty, A., Ranu, S., Bagchi, A., 2022. FairFoody: Bringing in fairness in food delivery, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 11900–11907. TODO verify pages. source: plan anchor “FairFoody AAAI 2022”.

Haferkamp, J., Ulmer, M.W., Ehmke, J.F., 2024. Heatmap-based decision support for repositioning in ride-sharing systems. Transportation Science 58, 110–130. Source: director (key); volume, number and pages added from the thesis entry.

Hildebrandt, F.D., Thomas, B.W., Ulmer, M.W., 2023. Opportunities for reinforcement learning in stochastic dynamic vehicle routing. Computers & Operations Research , 106071.

Lesmana, N.S., Zhang, X., Bei, X., 2019. Balancing efficiency and fairness in on-demand ridesourcing, in: Advances in Neural Information Processing Systems, pp. 5310–5320. Source: thesis.

Liao, W., Zhang, L., Wei, Z., 2020. Multi-objective green meal delivery routing problem based on a two-stage solution strategy. Journal of Cleaner Production 258, 120627. Source: thesis.

Liu, S., Luo, Z., 2023. On-demand delivery from stores: Dynamic dispatching and routing with random demand. Manufacturing & Service Operations Management 25, 595–612.

Martínez-Sykora, A., McLeod, F., Cherrett, T., Friday, A., 2024. Exploring fairness in food delivery routing and scheduling problems. Expert Systems with Applications 240, 122488. The “Southampton ESWA 2024” anchor of the plan. source: thesis.

Nair, A., Yadav, R., Gupta, A., Chakraborty, A., Ranu, S., Bagchi, A., 2022. Gigs with guarantees: Achieving fair wage for food delivery workers, in: Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI-22), pp. 5122–5128. The WORK4FOOD system. TODO verify title and pages. source: plan anchor “WORK4FOOD IJCAI 2022”.

Neria, G., Hildebrandt, F.D., Tzur, M., Ulmer, M.W., 2025. The restaurant meal delivery problem with ghost kitchens. Transportation Science 59, 433–450. Source: thesis.

Pillac, V., Gendreau, M., Guéret, C., Medaglia, A.L., 2013. A review of dynamic vehicle routing problems. European Journal of Operational Research 225, 1–11. Source: thesis.

Powell, W.B., 2019. A unified framework for stochastic optimization. European Journal of Operational Research 275, 795–821.

Powell, W.B., 2021. From reinforcement learning to optimal control: A unified framework for sequential decisions, in: Handbook of Reinforcement Learning and Control. Springer, pp. 29–74.

Quintero Rojas, S., 2020. Computational framework for solving the meal delivery routing problem. Ph.D. thesis. Department of Industrial Engineering, Universidad de los Andes. Bogotá, Colombia. Mdrp-sim. TODO confirm thesis type (MSc). source: thesis.

Reyes, D., Erera, A., Savelsbergh, M., Sahasrabudhe, S., O'Neil, R., 2018. The meal delivery routing problem. Optimization Online. Instance library mdrplib (Grubhub), https://github.com/grubhub/mdrplib. source: thesis.

Simoni, M.D., Winkenbach, M., 2023. Crowdsourced on-demand food delivery: An order batching and assignment algorithm. Transportation Research Part C: Emerging Technologies 149, 104055. Source: thesis.

Sobhanan, A., Charkhgard, H., Dayarian, I., 2024. Equity-driven workload allocation for crowdsourced last-mile delivery. SSRN preprint. Source: thesis.

Soeffker, N., Ulmer, M.W., Mattfeld, D.C., 2022. Stochastic dynamic vehicle routing in the light of prescriptive analytics: A review. European Journal of Operational Research 298, 801–820.

Steever, Z., Karwan, M.H., Murray, C.C., 2019. Dynamic courier routing for a food delivery service. Computers & Operations Research 107, 173–188. Source: thesis.

Stein, J., Hildebrandt, F.D., Ulmer, M.W., Thomas, B.W., 2025. Learning state-dependent policy parametrizations for dynamic technician routing with rework. Transportation Science .

TODO authors, 2025. TODO title: reinforcement learning for fair courier assignment in food delivery. `arXiv:2501.05808`. arXiv:2501.05808. TODO fetch authors and title. source: plan anchor "RL arXiv 2501.05808".

Ulmer, M.W., Goodson, J.C., Mattfeld, D.C., Thomas, B.W., 2020. On modeling stochastic dynamic vehicle routing problems. EURO Journal on Transportation and Logistics 9, 100008.

Ulmer, M.W., Thomas, B.W., 2020. Meso-parametric value function approximation for dynamic customer acceptances in delivery routing. European Journal of Operational Research 285, 183–195.

Ulmer, M.W., Thomas, B.W., Campbell, A.M., Woyak, N., 2021. The restaurant meal delivery problem: Dynamic pickup and delivery with deadlines and random ready times. Transportation Science 55, 75–100.

Wen, M., Cordeau, J.F., Laporte, G., Larsen, J., 2010. The dynamic multi-period vehicle routing problem. Computers & Operations Research 37, 1615–1623. Source: thesis; same key as the director's ReferencesAppx.bib.

Yıldız, B., Savelsbergh, M.W.P., 2019. Provably high-quality solutions for the meal delivery routing problem. Transportation Science 53, 1372–1388. Source: thesis.

Yitzhaki, S., 1979. Relative deprivation and the Gini coefficient. The Quarterly Journal of Economics 93, 321–324. Mean-absolute-difference form of the Gini used in eq:gini.

Appendix

In this Appendix, we present a more detailed literature review, an illustrative example of the decision process, the complete formulation of the per-round programme of **MIP-Fair**, and additional computational results.

Appendix A. Detailed literature review

This section reviews the literature related to the three components of the problem: the operational MDRP, fairness and compensation in delivery platforms, and courier behaviour. Table 1 in the main body provides a systematic summary of their similarities and differences relative to our work. [TODO: Move the fuller review of the thesis here (operational, economic and behavioural blocks), one subsection each, closing with the gap; switch citations to `citeAppx` (multibib, commented out in the preamble) and add ReferencesAppx.bib so that the appendix references do not count against the page limit, as in the director's manuscripts.]

Appendix A.1. Operational block

Reyes et al. (2018) defined the MDRP as a rolling-horizon, dynamic-deterministic problem and released the Grubhub instance library; Yıldız and Savelsbergh (2019) obtained provably high-quality solutions for a deterministic variant; Steever et al. (2019) revise assignments on every event; Ulmer et al. (2021) anticipate random ready times with approximate dynamic programming; and later contributions address shared logistics (Bi et al. 2023), courier operating regions (Auad et al. 2023), batching for crowdsourced fleets (Simoni and Winkenbach 2023), ghost kitchens (Neria et al. 2025) and on-demand delivery from stores (Liu and Luo 2023). In all of these, acceptance is perfect or exogenous. [TODO: expand.]

Appendix A.2. Economic block

Most operational models pay a fixed amount per delivery or minimise distance as a cost proxy. Reyes et al. (2018) and Yıldız and Savelsbergh (2019) include a shift-level guaranteed minimum set by the company; Wen et al. (2010) balance workload and income opportunities within a multi-period routing model; Lesmana et al. (2019) trace the efficiency–fairness frontier of ride-sourcing; Bajec and Tuljak-Suban (2022) use Shapley-value incentives; Sobhanan et al. (2024) give equity–cost frontiers; Gupta et al. (2022), Nair et al. (2022) and Martínez-Sykora et al. (2024) bring fairness to food delivery. None integrates acceptance-dependent pay, a real tariff and an intertemporal guarantee within a rolling-horizon routing problem. [TODO: expand.]

Appendix A.3. Behavioural block

Liao et al. (2020) balance workload and argue that equitable distribution is an incentive; reinforcement-learning approaches (Ahamed et al. 2021, Bozanta et al. 2022) include availability in the state but keep acceptance exogenous; Haferkamp et al. (2024) address repositioning; Quintero Rojas (2020) makes acceptance depend on distance. The fair-machine-learning literature shows that short- and long-horizon fairness diverge (Corbett-Davies et al. 2023). [TODO: expand; add TODO authors (2025).]

Appendix B. Illustrative example and algorithmic details

In this section, we first provide an illustrative numerical example connecting the sequential decision process to the dispatching decisions faced by the platform. We then present the complete per-round formulation of **MIP-Fair** and the tuning of the policy parameters.

Appendix B.1. Illustrative numerical example

We use a two-courier example to connect the components of Section 3.2 with the dispatching decisions and to show why a policy that is fair within every window can be unfair at the end of the day. Couriers A and B are connected from 12:00 to 13:00, so $\mathrm{act}_A(T) = \mathrm{act}_B(T) = 60$ minutes whatever is dispatched; the tariff is flat at 2,200 COP per order; B is slightly farther from the store than A. Four routes become available during the hour: at 12:00 a bundle of three orders from one store (pay 6,600 COP, duration 30 minutes including service), and at 12:30, 12:40 and 12:50 one singleton each (pay 2,200 COP, duration 8 minutes, so that the courier who takes one is free again before the next window). Coverage is the first priority of every policy, so all four routes are dispatched in every case; the question is to whom.

At $t =$ 12:00 the state is S_0 with both couriers idle and $h_A(0) = h_B(0) = (0, 0, 0)$. Both the per-window policy and the history-aware policy dispatch the bundle to A, the nearer courier, since nothing distinguishes the two couriers yet. At $t =$ 12:30 the state contains $h_A = (6{,}600, 3, 30)$ and $h_B = (0, 0, 30)$, that is, hourly earnings so far of $w_A(t) = 13{,}200$ COP/h and $w_B(t) = 0$, and one singleton is available. Consider two decisions. The per-window policy balances the pay offered within the window among the free couriers: with a single route, the window statistic is the same whichever courier receives it (one courier is offered 2,200 and the other 0), so the tie is broken by distance and the singleton goes to A. The same happens at 12:40 and 12:50. The end-of-day hourly earnings are $w_A = 6{,}600 + 3 \times 2{,}200 = 13{,}200$ COP/h and $w_B = 0$, and by Eq. (7)

$$F_{\mathrm{Gini}} = \frac{2 \times |13{,}200 - 0|}{2 \times 2 \times 13{,}200} = 0.5.$$

The history-aware policy reads $\mathcal{H}_t$ at 12:30: $\mathrm{deficit}_B = 1$ and $\mathrm{deficit}_A = 0$ in Eq. (11), or equivalently $w_B(s) = 3{,}474$ and $w_A(s) = 13{,}895$ COP/h in Eq. (12) (projected to the end of the window), so

the singleton goes to B, and likewise at 12:40 and 12:50. The end-of-day hourly earnings are $w_A = w_B = 6{,}600$ COP/h and $F_{\text{Gini}} = 0$, with the same coverage (six orders), the same active time, and a difference in travel of a few hundred metres.

This example highlights two points. First, the per-window statistic cannot distinguish the two decisions: within each window after 12:00 the pay offered is $(2{,}200, 0)$ whichever courier is chosen, so a policy that is fair within the window, such as the **Hierarchical** benchmark, is indifferent between an end-of-day Gini coefficient of 0.5 and one of 0. Second, what distinguishes them is the component $\mathcal{H}_t$ of the state, which records that A has already earned 6,600 COP during the same active time in which B has earned nothing. The effect compounds when couriers connect at different times, when routes dispatched in different windows differ in value, and when some couriers are systematically nearer to the stores, which is why the **Hierarchical** policy raises the end-of-day Gini coefficient on real days (Section 6.1).

Appendix B.2. Per-round formulation of MIP-Fair

At round t, let C_t be the set of connected couriers, $C_t^{\text{free}} \subseteq C_t$ the free ones, K_t the candidate pairs and $\mathcal{O}_t^{\text{ready}}$ the ready orders. Let $x_{cs} \in \{0, 1\}$ for $(c, s) \in K_t$, $y_o \in \{0, 1\}$ for $o \in \mathcal{O}_t^{\text{ready}}$, $w_c \geq 0$ for $c \in C_t$, $d_{ij} \geq 0$ for free–free pairs $i < j$, $\text{late}_o \geq 0$ and $v \geq 0$. The three stages of Eq. (14) are solved over the following constraints:

$$\sum_{(c,s)\in K_t : o\in s} x_{cs} \geq y_o, \qquad \forall o \tag{B.1}$$

$$\sum_{(c,s)\in K_t : o\in s} x_{cs} \leq 1, \qquad \forall o \tag{B.2}$$

$$\sum_{s:(c,s)\in K_t} x_{cs} \leq 1, \qquad \forall c \tag{B.3}$$

$$w_c = w_c^0\Big(1 - \sum_s x_{cs}\Big) + \sum_s x_{cs}\, w_c(s), \qquad \forall c \tag{B.4}$$

$$w_c = w_c^0, \qquad \forall c \tag{B.5}$$

$$d_{ij} \geq w_i - w_j, \quad d_{ij} \geq w_j - w_i, \qquad \forall i \tag{B.6}$$

$$\sum_{i<j\in C_t^{\text{free}}} d_{ij} + \sum_{i\in C_t^{\text{free}},\, j\notin C_t^{\text{free}}} \Big[\big(1-\textstyle\sum_s x_{is}\big)|w_i^0 - w_j^0| + \sum_s x_{is}|w_i(s) - w_j^0|\Big] + \sum_{i<j\notin C_t^{\text{free}}} |w_i^0 - w_j^0| - \alpha\, n \sum_{c\in C_t} w_c \leq n\, W_{\text{ref}}\, v \tag{B.7}$$

$$\text{late}_o \geq \sum_{(c,s)\in K_t : o\in s} x_{cs}\ \max(0, \hat{\tau}_{\text{drop}}(c,s,o) - d_o), \qquad \forall o \tag{B.8}$$

$$x_{cs} \in \{0,1\},\ y_o \in \{0,1\},\ d_{ij}, w_c, \text{late}_o, v \geq 0. \tag{B.9}$$

Eq. (B.1) links coverage to dispatched routes and Eqs. (B.2) - (B.3) ensure that each order is in at most one dispatched route and each courier receives at most one. Eqs. (B.4) - (B.5) give the projected hourly earnings, linear in x for free couriers and fixed for the others. Eq. (B.6) defines the pairwise differences for free–free pairs only, and Eq. (B.7) is the Gini constraint of Eq. (13) with the free–fixed block written out as linear terms and the fixed–fixed block as a constant; $n = |C_t|$ and W_{ref} is the mean of w_c^0 over C_t (or w^* when that mean is zero). Eq. (B.8) computes the projected lateness of each order under its dispatched route, and Eq. (B.9) defines the domains. Stage 1 maximises $\sum_o y_o$; stage 2 fixes that value and minimises v; stage 3 fixes v^* and minimises $\lambda_{\text{tt}} \sum_{(c,s)} \tau_s x_{cs} + \mu \sum_o \text{late}_o$. We use $\lambda_{\text{tt}} = 0.01$ per minute of travel and $\mu = 0.02$ per minute of lateness, the tie-breaking weights of the **ILP** and **Hierarchical** policies.

Appendix B.3. Tuning of the policy parameters

Table C.11 in Appendix C.4 reports the two tuning ablations on the 24 instances: the BRKGA with population 20 and 20 generations leaves coverage unchanged and raises the end-of-day Gini coefficient by 0.007 at almost three times the CPU time, and the **MIP-Fair** prefilter with $k = 25$ lowers it by 0.039 and raises coverage by 0.007 at eight times the CPU time (31 minutes per day on average, 227 at most, and beyond the four-hour timeout on the largest Bogotá day on four of the five seeds); the deficit-aware prefilter of Section 6.1 obtains more than twice that reduction for the same order of CPU time. The settings of Section 5.5 are therefore kept. In the pilot, the BRKGA-type policies take 9–10 seconds for a full day of Pereira and 30–35 seconds for a two-hour window of Bogotá, the **Hierarchical** policy 16 and 39 seconds, and **MIP-Fair** with $k = 8$ 36 seconds and 4.5 minutes, respectively; a BRKGA with population 20 and 20 generations did not change coverage on Pereira.

Appendix C. Additional results of the experiments

In this section, we provide the detail that complements the main results: the instances, parameters and factors of the study, the results by city, the compliance and the earnings by population, the ablations of X3, the Gini coefficient under the sensitivity settings together with the regime loss and the choice of α, and the emissions diagnostic.

Appendix C.1. Instances, parameters and factors

Table C.6 lists the 24 instances described in Section 5.1, Table C.5 the parameters of the simulation with their values, sources and evidence grade, and Table C.7 the factors and levels of the computational study of Section 5.3.

Parameter	Value	Source	Grade
Acceptance threshold θ	903 COP/km (95% [615; 1,242])	interviews, 32 decisions / 11 couriers	C
Acceptance slope b	2.89 (95% [2.02; 5.23])	interviews	C
Acceptance regime	manual / auto-accept	interviews (A), survey 64%	A
Tariff π_0	2,200 COP per order, flat	survey ($n = 132$)	B
Minimum wage w^*	4,367.91 COP/h	statutory 2021 / (48 h × 4.33)	–
In-store service	20 min (base), 60 min (stress)	survey / interviews	B / C
Drop-off service	4 min	public instances	–
Decision delay	16 ± 8 s	interviews (IQR 9–28 s)	C
Speeds (bike/moto/walk/car)	15 / 25 / 5 / 20 km/h	assumption	–
Circuity κ	1.3 (1.0 sensitivity)	assumption	–
Hard deadline $\Delta^{\max}$	promised time + 30 min	assumption (interviews)	–
Route capacity Q	3	simulator default	C
Window	2 min	platform practice	–
Emission factors	moto 0.114, car 0.167, bike/walk 0 kg CO_2e/km	Department for Energy Security and Net Zero and Department for Environment, Food & Rural Affairs (UK) (2023)	–

Table C.5: Parameters, values, sources and evidence grade

Appendix C.2. Results by city

Table C.8 repeats the manual-regime block of Table 2 for each of the six cities, with the compliance with $\alpha = 0.40$ as the last column. The central result holds city by city: **Greedy** is between 0.275 (Medellín) and 0.323 (Pereira) on the end-of-day Gini coefficient over the eligible couriers and meets $\alpha = 0.40$ on every day except three in Pereira, while the six order-centric policies

Inst.	City	Orders	Couriers	Stores	Orders/courier	Vehicle mix (full city, %: bike/moto/walk/car)
0	Bogotá	10,215	2,650	926	3.9	45/39/9/7
6	Bogotá	9,743	2,489	997	3.9	43/39/10/8
12	Bogotá	5,618	1,967	859	2.9	44/41/6/9
18	Bogotá	5,875	1,931	912	3.0	39/39/11/11
2	Medellín	6,169	1,692	602	3.6	16/72/6/7
8	Medellín	5,778	1,773	628	3.3	15/71/7/7
14	Medellín	3,953	1,244	614	3.2	15/72/5/8
20	Medellín	3,918	1,259	587	3.1	15/70/7/8
5	Barranquilla	7,773	1,442	470	5.4	21/64/4/10
11	Barranquilla	7,116	1,532	474	4.6	22/63/4/11
17	Barranquilla	3,606	1,168	4,014[a]	3.1	19/66/3/13
23	Barranquilla	3,612	1,046	431	3.5	20/59/5/15
7	Cali	1,813	745	269	2.4	21/72/2/5
13	Cali	1,262	488	260	2.6	21/73/2/5
19	Cali	1,292	476	264	2.7	22/71/2/5
1	Bucaramanga	1,851	574	216	3.2	9/89/1/1
3	Bucaramanga	1,851	574	216	3.2	9/89/1/1
9	Bucaramanga	1,527	610	218	2.5	9/89/1/1
15	Bucaramanga	976	536	202	1.8	7/90/1/1
21	Bucaramanga	948	498	204	1.9	9/87/2/1
4	Pereira	568	253	135	2.2	5/91/2/2
10	Pereira	498	277	140	1.8	6/90/2/2
16	Pereira	342	268	401[a]	1.3	5/90/2/3
22	Pereira	374	246	122	1.5	6/89/3/3

[a] One store record per order in the export; does not affect the simulation. Instances 1 and 3 are different days with identical counts.

Table C.6: The 24 instances: orders, couriers and stores in the 3-km zone, and vehicle mix of the full city

Factor	Levels	Acts on
Policy	**Greedy**, **ILP**, **BRKGA**, **Hierarchical**; **BRKGA-Fair** $w \in \{5, 10, 20, 40\}$ min; **MIP-Fair** $\alpha \in \{0.20, 0.25, 0.30, 0.35, 0.40, 1.00\}$; **Pay-BRKGA** $\lambda_{\text{pay}} \in \{0.1, 0.2, 0.5, 1.0\}$	round
Acceptance regime	manual, auto-accept	Eq. (4)
Logit threshold θ (X2, manual regime)	615 / 903 / 1,242 COP/km, slope fixed at 2.89	Eq. (4)
Instance	24 (six cities), 3-km zone	block
Seed	10 in X1, five in X2 and X3, the same in every cell	acceptance, delays, BRKGA

Table C.7: Factors and levels of the computational study

are between 0.448 (**ILP**, Bogotá and Medellín) and 0.637 (**BRKGA**, Bucaramanga and Pereira) and meet it on none. The guarantee is therefore uniformly within reach of the courier-centric rule, and the question of where it is cheap or expensive becomes the question of where the order-centric policies fall furthest from it. They fall furthest in the two smallest fleets, Bucaramanga and Pereira (0.56–0.64, with only 59–73% of the eligible couriers active), and closest in Bogotá and Medellín (0.45–0.51, 79–86% active), where the fleet is large relative to demand and the best-placed courier changes more often. The coverage that **Greedy** gives up with respect to the best policy of the city is between 0.8 points (Bogotá) and 1.5 points (Cali), and its extra lateness between 0.9 (Pereira, against **Hierarchical**) and 2.3 minutes (Bogotá); the pay lever of **Pay-BRKGA** cuts the bill most where the fleet is small and routes are long (Pereira, 19%; Bucaramanga, 15%) and least in Bogotá and Medellín (7–9%). The second-generation variants with $k_d = 25$ (five seeds) keep the ordering: under the manual regime they reach 0.37–0.38 in Barranquilla, Bogotá and Medellín, 0.45–0.47 in Cali and 0.49–0.55 in Bucaramanga and Pereira, so the gap to **Greedy** that remains is again largest in the two smallest fleets.

Appendix C.3. Compliance and earnings by population

Table C.9 reports the compliance with the end-of-day guarantee by policy and regime, Table C.10 reports the end-of-day Gini coefficient by population discussed in Section 6.1, and Figure C.6 the distribution of end-of-day hourly earnings by policy discussed in Section 6.3.

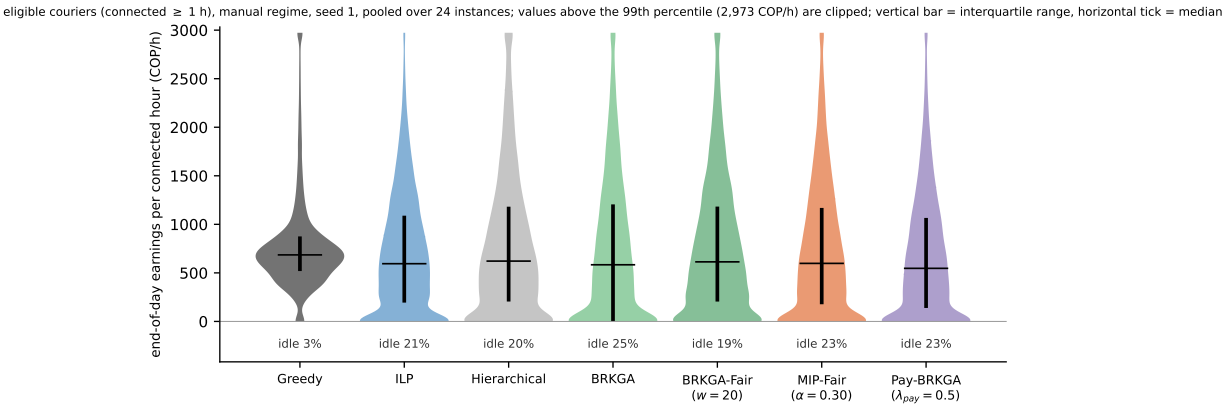


Figure C.6: Distribution of end-of-day hourly earnings over the eligible couriers by policy (manual regime, seed 1, 24 instances pooled; values above the 99th percentile clipped; bar = interquartile range, tick = median)

Appendix C.4. Ablations

Table C.11 reports the ablations of the supplement of X3, each paired with its base configuration on the same (instance, seed) pairs under the manual regime, together with the $k_d = 50$ rows of the second generation under both regimes. The table uses the five seeds of X3. The **MIP-Fair** rows with the free-only scope, the total-earnings basis and $k = 25$ have 115–116 pairs because their

Policy	Coverage	End-of-day Gini F_{Gini}	Active eligible	Lateness (min)	Payment (% of tariff)	$F_{\text{Gini}} \leq 0.40$ (% of days)
Barranquilla (4 instances)						
Greedy	0.692	**0.304**	**0.98**	20.0	100	100
ILP	0.670	0.470	0.80	19.8	100	0
Hierarchical	0.700	0.491	0.80	**18.2**	100	0
BRKGA	0.701	0.513	0.76	18.7	100	0
BRKGA-Fair ($w = 20$)	0.701	0.481	0.82	18.8	100	0
MIP-Fair ($\alpha = 0.30$)	0.690	0.503	0.78	19.5	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.690	0.499	0.78	19.0	**92**	0
Bogotá (4 instances)						
Greedy	0.811	**0.282**	**0.96**	20.1	100	100
ILP	0.799	0.448	0.85	19.6	100	0
Hierarchical	0.815	0.503	0.81	**17.8**	100	0
BRKGA	0.817	0.509	0.79	18.5	100	0
BRKGA-Fair ($w = 20$)	**0.819**	0.474	0.86	18.7	100	0
MIP-Fair ($\alpha = 0.30$)	0.808	0.484	0.82	19.4	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.809	0.493	0.81	18.8	**93**	0
Bucaramanga (5 instances)						
Greedy	0.730	**0.306**	**0.97**	19.6	100	100
ILP	0.714	0.593	0.63	19.1	100	0
Hierarchical	0.743	0.579	0.71	**18.3**	100	0
BRKGA	0.743	0.637	0.59	18.6	100	0
BRKGA-Fair ($w = 20$)	0.743	0.607	0.65	18.6	100	0
MIP-Fair ($\alpha = 0.30$)	0.741	0.625	0.60	18.7	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.726	0.613	0.61	18.8	**85**	0
Cali (3 instances)						
Greedy	0.748	**0.322**	**0.96**	20.7	100	100
ILP	0.727	0.530	0.70	19.9	100	0
Hierarchical	0.762	0.563	0.71	**18.5**	100	0
BRKGA	0.762	0.591	0.63	18.8	100	0
BRKGA-Fair ($w = 20$)	**0.763**	0.562	0.69	18.9	100	0
MIP-Fair ($\alpha = 0.30$)	0.759	0.554	0.68	19.2	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.743	0.564	0.66	19.2	**86**	0
Medellín (4 instances)						
Greedy	0.812	**0.275**	**0.99**	18.7	100	100
ILP	0.797	0.448	0.82	18.0	100	0
Hierarchical	0.823	0.457	0.85	**17.0**	100	0
BRKGA	0.824	0.490	0.79	17.4	100	0
BRKGA-Fair ($w = 20$)	0.824	0.467	0.85	17.5	100	0
MIP-Fair ($\alpha = 0.30$)	0.820	0.474	0.82	17.7	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.815	0.475	0.81	17.6	**91**	0
Pereira (4 instances)						
Greedy	0.762	**0.323**	**0.95**	20.3	100	85
ILP	0.723	0.619	0.59	20.3	100	0
Hierarchical	**0.776**	0.564	0.73	**18.8**	100	0
BRKGA	0.775	0.637	0.61	19.1	100	0
BRKGA-Fair ($w = 20$)	0.774	0.619	0.65	19.1	100	0
MIP-Fair ($\alpha = 0.30$)	0.770	0.610	0.63	19.4	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.742	0.621	0.61	19.4	**81**	0

Table C.8: Results by city, manual regime: mean over the instances of the city (ten seeds averaged within each instance) for the configurations of Table 2; the last column is the share of (instance, seed) pairs with $F_{\text{Gini}} \leq 0.40$. Best value per column and city in bold.

Policy	Manual					Auto-accept				
	$\alpha = 0.30$	0.35	0.40	0.45	0.50	$\alpha = 0.30$	0.35	0.40	0.45	0.50
Greedy	**50**	**90**	**98**	**100**	**100**	**54**	**96**	**100**	**100**	**100**
ILP	0	0	0	18	52	0	0	4	46	58
Hierarchical	0	0	0	4	34	0	0	0	5	42
BRKGA	0	0	0	0	24	0	0	0	0	20
BRKGA-Fair ($w = 20$)	0	0	0	5	45	0	0	0	5	45
MIP-Fair ($\alpha = 0.30$)	0	0	0	4	40	0	0	0	8	46
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0	0	0	3	37	0	0	0	4	45
Second generation (five seeds)										
BRKGA-Fair ($w = 40$, $k_d = 25$)	0	8	49	62	72	2	20	56	67	81
MIP-Fair ($\alpha = 0.30$, $k_d = 25$)	0	8	40	61	82	0	12	63	83	96

Table C.9: Compliance with the end-of-day guarantee: % of (instance, seed) pairs with $F_{\text{Gini}} \leq \alpha$, by regime.

Policy	Eligible couriers (connected ≥ 1 h) F_{Gini}, idle at zero	active only	Active couriers $F^{\text{act}}_{\text{Gini}}$	Active share of eligible	First quartile (COP/h, eligible)
Greedy	**0.301**	**0.279**	0.509	**0.97**	**481**
ILP	0.521	0.345	**0.414**	0.73	130
Hierarchical	0.527	0.387	0.530	0.77	146
BRKGA	0.565	0.376	0.488	0.70	86
BRKGA-Fair ($w = 20$)	0.537	0.385	0.500	0.75	130
MIP-Fair ($\alpha = 0.30$)	0.544	0.366	0.460	0.72	105
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.546	0.364	0.457	0.71	97

Table C.10: The end-of-day Gini coefficient of hourly earnings by population, manual regime, mean over the 24 instances: the statistic of Eq. (7), its restriction to the eligible couriers that delivered, and the active-courier statistic $F^{\text{act}}_{\text{Gini}}$, which includes the couriers connected for less than one hour.

runs on the largest Bogotá day (instance 0) did not finish within the 240-minute timeout on four or five of the seeds, and the equity-first row has 27 pairs because the configuration was withdrawn from the grid after those runs (98 minutes per run for a policy that the pilot had already shown to be degenerate). The statistic to read is the paired difference with its 95% interval; the column "worse/better" counts the pairs in which the ablation ends more than 0.005 above or below its base in F_{Gini}.

Ablation (base)	n	Coverage mean	Coverage Δ base	End-of-day Gini F_{Gini} mean	End-of-day Gini F_{Gini} Δ base [95% CI]	Worse / better	Active eligible	Gini $F^{\text{act}}_{\text{Gini}}$	CPU time min (max)	CPU $\times$ base
Manual regime										
BRKGA decoder $\text{score}_{\text{finish}}$ (base: finish per order)	120	0.769	−0.001	0.558	−0.007 [-0.008, -0.005]	15/75	0.70	0.372	3.3 (17)	1.4
BRKGA 20×20 (base: 10×10)	120	0.769	−0.000	0.571	+0.007 [+0.005, +0.008]	76/8	0.69	0.380	6.2 (27)	2.7
BRKGA-Fair ($w = 10$) total-earnings basis (base: hourly)	120	0.770	+0.000	0.534	−0.005 [-0.006, -0.003]	12/55	0.75	0.380	3.3 (17)	1.4
MIP-Fair Gini over free couriers only (base: all connected)	115	0.769	+0.006	0.498	−0.050 [-0.059, -0.041]	7/97	0.75	0.330	32.7 (183)	8.6
MIP-Fair total-earnings basis (base: hourly)	116	0.770	+0.007	0.515	−0.032 [-0.043, -0.022]	31/78	0.73	0.336	33.1 (240)	8.3
MIP-Fair equity-first priority (base: coverage first)	27	0.285	−0.469	0.153	−0.407 [-0.496, -0.317]	1/26	0.41	0.133	97.6 (227)	36.3
MIP-Fair prefilter $k = 25$ (base: $k = 8$)	115	0.770	+0.007	0.508	−0.039 [-0.047, -0.032]	9/98	0.74	0.334	30.6 (227)	8.0
MIP-Fair coverage slack $\varepsilon = 0.02$ (base: 0)	120	0.763	−0.001	0.543	−0.001 [-0.003, +0.000]	26/39	0.72	0.363	6.6 (35)	1.5
MIP-Fair coverage slack $\varepsilon = 0.05$ (base: 0)	120	0.763	−0.000	0.544	−0.000 [-0.002, +0.001]	27/24	0.72	0.365	6.6 (35)	1.5
BRKGA-Fair ($w = 40$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.768	−0.001	0.404	−0.033 [-0.036, -0.030]	0/118	0.88	0.323	6.4 (34)	1.3
MIP-Fair ($\alpha = 0.30$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.763	−0.004	0.412	−0.023 [-0.028, -0.018]	17/89	0.87	0.325	51.1 (234)	2.2
Auto-accept regime										
BRKGA-Fair ($w = 40$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.786	+0.000	0.352	−0.064 [-0.068, -0.061]	0/120	0.91	0.286	5.5 (30)	1.3
MIP-Fair ($\alpha = 0.30$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.786	+0.000	0.398	−0.002 [-0.007, +0.002]	44/67	0.88	0.316	22.4 (104)	2.0

Table C.11: Ablations of the supplement of X3, paired with the base configuration of each row on the same (instance, seed) pairs: number of pairs, coverage and F_{Gini} of the ablation with the paired difference with respect to the base, number of pairs worse / better than the base by more than 0.005 in F_{Gini}, active share of the eligible couriers, Gini coefficient over the active couriers, CPU time per run (mean and maximum, minutes) and its ratio to the base. Manual regime, five seeds of X3 (1,453 runs); the $k_d = 50$ rows also under auto-accept.

BRKGA decoder and size. Scoring a chromosome by the finishing time instead of the finishing time per order lowers F_{Gini} by 0.007 and coverage by 0.001 at 1.4 times the CPU time. A population of 20 with 20 generations leaves coverage unchanged, raises F_{Gini} by 0.007 (worse in 76 of the 120 pairs) and costs almost three times the CPU time: the round objective is a coverage objective, and a longer search improves a quantity that the orders whose deadline can no longer be met already saturate while doing nothing for the end-of-day statistic. The 10×10 setting and the per-order score are retained.

*Scope and basis of the **MIP-Fair** constraint, and prefilter k.* Three ablations tighten the round constraint of **MIP-Fair**: the Gini over the free couriers only, the total-earnings basis and the prefilter with $k = 25$. The three lower F_{Gini} by 0.032–0.050 and raise coverage by 0.006–0.007, with 78–98 of the pairs better than the base. The expectation that the free-only scope would raise the end-of-day Gini because the free couriers are a biased subset is not borne out: F_{Gini} falls by 0.050 and the active-courier Gini from 0.365 to 0.330. None of the three reaches the second-generation variant with $k_d = 25$ ($F_{\text{Gini}} = 0.435$, Table 3), and all three cost eight to nine times the CPU time of the base (31–33 minutes per day on average and up to four hours on a Bogotá day), which is consistent with rounds that run to the 10-second solver limit once the constraint binds. A tighter round constraint is therefore not a substitute for the candidate set. The total-earnings basis in **BRKGA-Fair** moves F_{Gini} by 0.005: the weighted score reads the deficit only through the $K = 3$ candidates of the decoder, so the basis matters little, as the exposure basis of the second generation also showed.

Equity-first priority. The lexicographic order with equity before coverage, the first design of **MIP-Fair**, reproduces the collapse of the pilot on the 27 runs completed: coverage 0.285 against 0.754 for the base, a Gini of 0.153 obtained by not assigning (41% of the eligible couriers active, lateness ten minutes lower because few orders are served) and 36 times the CPU time. The paper's **MIP-Fair** keeps coverage first for this reason.

Coverage slack. The slack $\varepsilon \in \{0.02, 0.05\}$ leaves F_{Gini} and coverage within 0.001 of the base, with as many pairs worse as better; this is the first-generation counterpart of the ε result in Table 3, and confirms that the lexicographic priority was not what bound the Gini block.

$k_d = 50$. The last two rows of each block answer the question left open in Section 6.1: whether the half of the gap to **Greedy** that remains at $k_d = 25$ is a matter of candidates or of the one-route-per-courier structure of the round. For **BRKGA-Fair** it is a matter of candidates: F_{Gini} falls from 0.437 to 0.404 under the manual regime and from 0.416 to 0.352 under auto-accept, no pair is worse than at $k_d = 25$, coverage is unchanged, 55% and 73% of the days meet $\alpha = 0.40$, and the CPU time grows by a third (6 minutes per day on average, 34 at most). The k_d curve of the heuristic, $0.534 \to 0.437 \to 0.404$ (manual) and $0.536 \to 0.416 \to 0.352$ (auto-accept) for $k_d \in \{0, 25, 50\}$, is still descending. For **MIP-Fair** the curve flattens: $0.544 \to 0.465 \to 0.435 \to 0.412$ (manual) and $0.534 \to 0.426 \to 0.401 \to 0.398$ (auto-accept) for $k_d \in \{0, 8, 25, 50\}$, the last step within noise under auto-accept and with a coverage loss of 0.004 under the manual regime, while the CPU time grows to 51 minutes per day on average and almost four hours on the largest Bogotá day. For the exact round model the structure of the round and the solver limit bind before the candidate set does.

Appendix C.5. Gini coefficient under the sensitivity settings and emissions

Table C.12 is the companion of Table 4 for the end-of-day Gini coefficient over the eligible couriers. The statistic moves less than coverage under every factor: the regime lowers it by 2% for the first-generation policies and by 5–8% for the second-generation ones, whose deficit-aware prefilter reaches more idle couriers when every offer is accepted; the threshold moves it by at most 3.8%; α leaves it unchanged in the first generation; the pilot did not record the eligible-courier statistic at circuity 1.0.

	Relative difference of F_{Gini} over the base-case setting (in %)									
Policy\Setting	**Regime**		**Threshold θ (COP/km)**			**Tolerated inequality α**			**Circuity**	
	manual	**auto**	**615**	**903**	**1,242**	**0.20**	**0.30**	**0.40**	**1.0**	**1.3**
Greedy	0.0	−1.7	−0.7	0.0	+3.5	–	–	–	–	0.0
Hierarchical	0.0	−2.2	−1.2	0.0	+2.2	–	–	–	–	0.0
MIP-Fair ($\alpha = 0.30$)	0.0	−1.9	–	0.0	–	0.0	0.0	+0.1	–	0.0
BRKGA-Fair ($w = 40$, $k_d = 25$)	0.0	−4.7	−2.3	0.0	+3.8	–	–	–	–	0.0
MIP-Fair ($\alpha = 0.30$, $k_d = 25$)	0.0	−7.8	−2.8	0.0	+3.2	–	–	–	–	0.0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.0	−2.1	−1.4	0.0	+2.5	–	–	–	–	0.0

Table C.12: Sensitivity of the end-of-day Gini coefficient over the eligible couriers across operational settings: relative difference over the base case (manual regime, $\theta = 903$, $\alpha = 0.30$, circuity 1.3); sources as in Table 4

Figure C.7 shows the regime loss discussed in Section 6.4, and Table C.13 the three criteria for the tolerated inequality α and the policy that satisfies each at the least coverage cost.

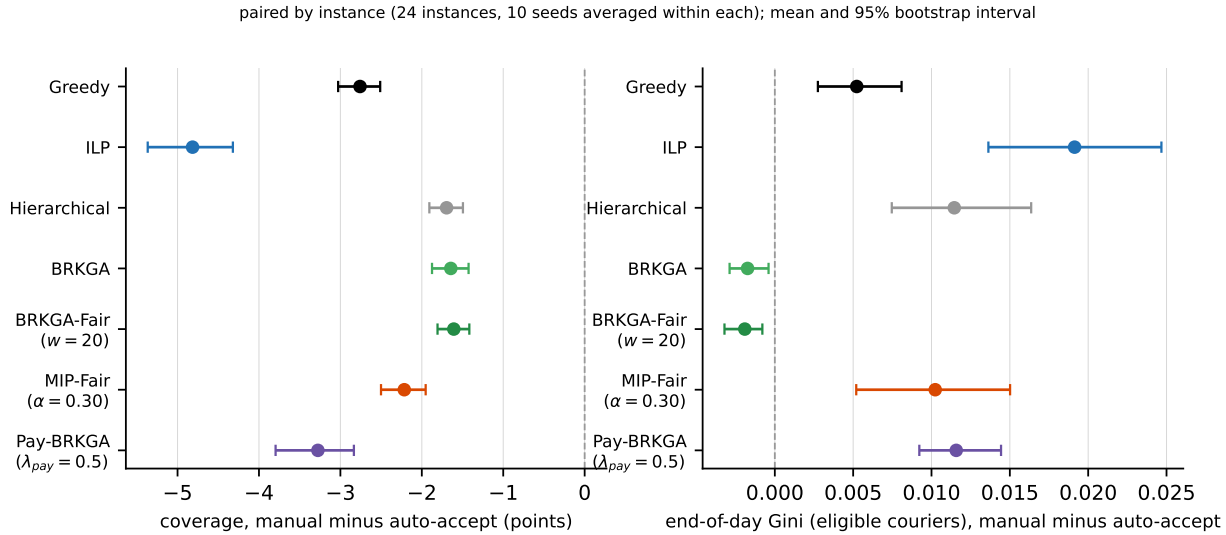


Figure C.7: Regime loss, manual minus auto-accept, paired by instance: coverage (left) and end-of-day Gini coefficient over the eligible couriers (right); mean and 95% bootstrap interval over the 24 instances, ten seeds averaged within each instance

Figure C.8 shows the emissions diagnostic discussed in Section 6.4: kg CO_2e per delivered order by policy and city (left) and the share of the distance driven by each vehicle type by policy (right),

Criterion	α	Policy satisfying it at least cost	Coverage cost vs. Greedy (pts)
A: historical (corporate Gini)	0.30 (X1: 0.301 manual, 0.296 auto-accept; 0.27–0.32 by city)	**Greedy** only; no order-centric policy meets it on any day	0 (status quo)
B: inequality band	0.30 (anchors 0.26 / 0.39); upper bound 0.40	at 0.40: **BRKGA-Fair** ($w = 40$, $k_d = 25$) on 49% of the days, **MIP-Fair** ($k_d = 25$) on 40%	−1.1 / −0.9 (coverage gain)
C: Pareto knee	none: coverage flat within 0.3 points along α and along k_d	–	–

Table C.13: The tolerated inequality α under the three criteria and the recommended policy

from the per-courier distances of the 240 manual-regime runs of each policy and the emission factors of Table C.5. The numbers behind the figure are in the online supplement (F8_emissions.csv).

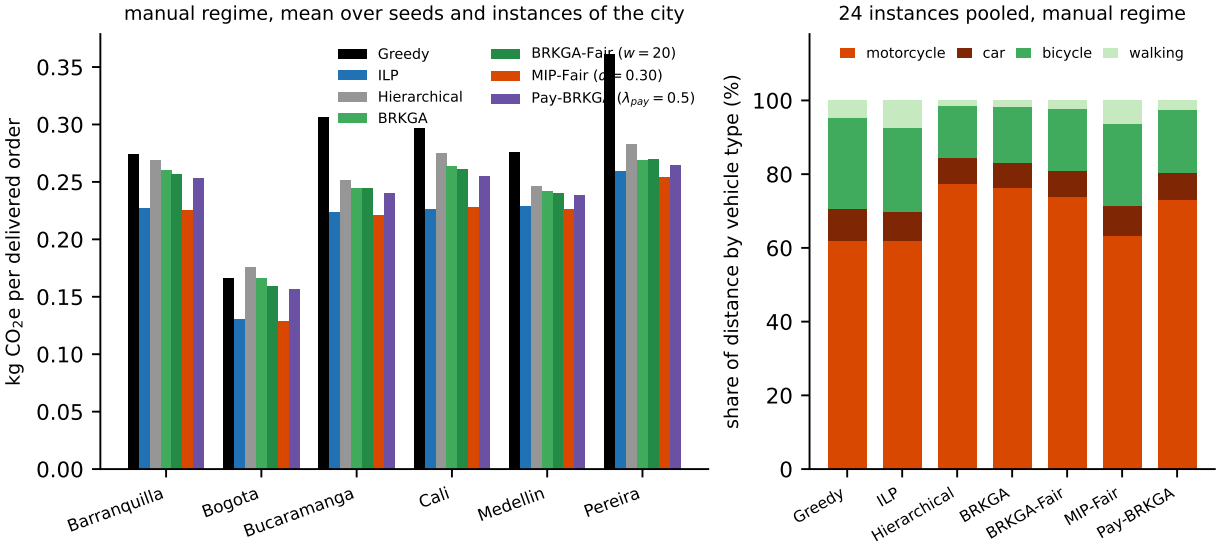


Figure C.8: Emissions diagnostic: kg CO_2e per delivered order by policy and city, manual regime, mean over the ten seeds and the instances of the city (left); share of the distance driven by each vehicle type, by policy, 24 instances pooled (right)